PREDICTING DENGUE FEVER PROGRESSION: PREDICTIVE ANALYTICS FOR PLATELET COUNT

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Submitted in Partial Fulfillment of the Requirements for the Award of the Degree of
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CERTIFICATE

This is to certify that the project titled

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Date:.....	Date:.....
Name and Address of the student	Name, Designation and Address of the Guide
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Project Proposal Report / Synopsis

for

Predicting Dengue Fever Progression: Predictive Analytics for Platelet Count

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Abbreviations

DENV - Dengue virus

DHF - Dengue haemorrhagic fever

DSS - Dengue shock syndrome

RBC - Red Blood Cells

PII - personally identifiable information

CSV - Comma separated values

XLSX - Microsoft Excel Open XML Spreadsheet

ML - Machine Learning

NaN - Not a number

EDA - Exploratory Data Analysis

LSTM - Long Short-Term Memory

RNN - Recurrent neural network

GRU - Gated Recurrent Unit

XGBoost - Extreme Gradient Boosting

RMSE - Root Mean Square Error

MAE - Mean Absolute Error

R^2 - R-squared

Predicting Dengue Fever Progression: Predictive Analytics for Platelet Count Data

Chapter 1

1.1 Overview / Background

Dengue fever is a rapidly spreading mosquito-borne viral disease that affects millions of people annually, particularly in Southeast Asia, the Western Pacific islands, Latin America, and Africa. Caused by the dengue virus (DENV), which exists in four serotypes (DENV-1 to DENV-4), the disease presents a wide range of clinical manifestations—from mild flu-like symptoms to life-threatening complications such as dengue haemorrhagic fever (DHF) and dengue shock syndrome (DSS). A particularly dangerous complication of severe dengue is thrombocytopenia, a significant drop in platelet count, which raises the risk of internal bleeding, organ failure, and even death.

Effective management of thrombocytopenia relies heavily on timely monitoring and intervention. However, traditional methods for tracking platelet levels are reactive and fail to provide predictive insight, often leading to delayed treatment. This gap between diagnostics and proactive care presents an opportunity to apply data science solutions for meaningful clinical impact.

The motivation behind this project stems from both the urgency of the healthcare challenge and the potential of data-driven technologies to address it. Rather than selecting a straightforward problem, this project was chosen precisely for its complexity and real-world relevance. It bridges the gap between clinical medicine and computational modeling by aiming to forecast platelet trends using machine learning and time series techniques. Through the analysis of historical clinical data, the goal is to develop a robust predictive model that can alert clinicians to imminent platelet drops—thus enabling timely intervention and potentially saving lives.

This project is not only a valuable academic exercise but also contributes to the broader mission of improving healthcare delivery in resource-constrained, dengue-endemic regions. It also

offers scope for future expansion, including integration with electronic health systems, real-time monitoring apps, or extension to other vector-borne diseases with similar clinical patterns. In doing so, the work aspires to enhance the role of artificial intelligence in public health and build scalable, efficient, and context-aware healthcare tools.

1.2 Problem-statement

Dengue fever is a mosquito-borne viral disease prevalent in tropical and subtropical regions, posing a significant public health concern. The disease manifests in a spectrum of symptoms, ranging from mild febrile illness to severe complications such as dengue haemorrhagic fever (DHF) and dengue shock syndrome (DSS). Thrombocytopenia, one of the critical complications of severe dengue, is a condition characterized by a drastic reduction in platelet count. A critically low platelet count can lead to life-threatening internal and external bleeding, requiring immediate medical attention. Therefore, continuous monitoring and timely intervention are essential to prevent severe outcomes in affected patients.

Current clinical practices primarily involve reactive monitoring of platelet levels through periodic laboratory tests. These methods, although effective, often result in delayed interventions due to their inability to predict drastic changes in platelet count. As a result, healthcare providers face challenges in administering timely treatments, leading to increased morbidity and mortality among dengue patients.

A data-driven approach to forecasting platelet counts can significantly enhance patient care by enabling early prediction of thrombocytopenia. By leveraging historical platelet count data, machine learning, and time series forecasting models, clinicians can anticipate severe declines in platelet levels and implement pre-emptive interventions. Such a predictive model would aid in risk stratification, allowing for timely administration of supportive treatments such as platelet transfusions, thereby reducing the risk of complications.

This research aims to develop a platelet count forecasting model using past patient data to predict future trends in platelet levels. The model will provide healthcare professionals with a proactive tool to enhance decision-making and optimize resource allocation. By integrating data

science methodologies into clinical practice, this study seeks to bridge the gap between reactive and predictive healthcare, ultimately improving patient outcomes in dengue management

1.3 Project Objectives

The primary objective of this research is to develop a predictive model for forecasting platelet counts in dengue patients using historical data. By employing data-driven methodologies, this study aims to enhance early detection and proactive management of thrombocytopenia, thereby reducing the risks associated with severe dengue complications.

The specific objectives of this study are as follows:

1. **To analyse historical platelet count trends** – Examining past patient data to identify patterns and fluctuations in platelet levels during the course of dengue fever.

2. **To develop a predictive model for platelet count forecasting** – Utilizing machine learning and time series forecasting techniques to predict future platelet levels based on past trends.

3. **To evaluate the performance of the proposed model** – Assessing the accuracy, reliability, and clinical applicability of the model by comparing its predictions with actual patient outcomes.

4. **To provide an early warning system for thrombocytopenia** – Enabling healthcare providers to anticipate severe platelet drops and administer timely interventions, thereby improving patient outcomes and reducing hospital burden.

By achieving these objectives, this research aims to bridge the gap between traditional reactive monitoring methods and proactive predictive healthcare, ultimately contributing to improved patient care and better management of dengue-related complications.

1.4 Organization details

St. Aloysius Institute of Management and Information Technology (AIMIT) is a private institute located in Mangalore, Karnataka, established in 2009. It is the new campus for MBA and IT programs offered by St Aloysius (Deemed to be University).

AIMIT is a member of XAJBS (Xavier Association of Jesuit Business Schools), India, which has prestigious B-schools like XLRI, XIMB, and LIBA as members. AIMIT is also a member of IJABS, the International Jesuit Association of Business Schools.

Vision:

"Empowering youth through excellence in education to shape a better future for humankind"

Mission:

St Aloysius Institutions of the Mangalore Jesuit Educational Society, inspired by the person and mission of Jesus Christ, and guided by the motto "Lucet et Ardet" meaning "Shine to Enkindle" commit themselves to spread the light of knowledge and wisdom and to kindle the ardour of faith that does justice by forming men and women for others, who are:

1. Academically accomplished,
2. Emotionally balanced,
3. Morally upright,
4. Socially responsible,
5. Ecologically sensitive
6. Professionally dedicated
7. so that they are a powerful force for the transformation of society.

Chapter 2: Literature Review

2.1 Introduction

Dengue fever, a mosquito-borne viral infection, remains a major public health challenge, particularly in tropical and subtropical regions where it causes substantial morbidity and mortality. Accurate and early prediction of disease severity and outbreak risk is crucial for improving patient outcomes and efficiently managing healthcare resources, especially in low-resource settings. Recent advancements in data science, particularly in machine learning (ML), deep learning, and statistical modeling, have opened new avenues for forecasting dengue incidence and classifying disease severity. The literature reviewed in this section spans diverse methodologies and focuses, including clinical and hematological predictors such as platelet indices and leukocyte counts, image-based ML diagnostics, and hybrid time series models like ARIMAX and autoregressive neural networks. Some studies also incorporate climate variables, vector indices, and unsupervised learning approaches to enhance prediction accuracy. Additionally, parallels are drawn with diseases such as Severe Fever with Thrombocytopenia Syndrome (SFTS), which shares clinical similarities with dengue. Collectively, these studies represent the current state of the art and contribute to a comprehensive understanding of how computational and clinical approaches can be leveraged to improve early diagnosis, severity classification, and outbreak forecasting of dengue fever.

2.2 Details of literature review

The growing burden of dengue fever, particularly in tropical and subtropical regions, has prompted extensive research into methods for improving diagnosis, severity assessment, and outbreak forecasting. Recent literature spans a variety of methodological approaches—from classical statistical analysis to advanced machine learning (ML), deep learning, and image-based diagnostics—across diverse clinical and geographical contexts.

Phakhounthong et al. (2018) conducted a retrospective study at the Angkor Hospital for Children in Cambodia, aiming to predict the severity of dengue in pediatric patients using clinical and laboratory indicators available at the time of admission. The study included 198

confirmed dengue cases and employed decision tree analysis to identify predictive factors. Key features evaluated included hematocrit, Glasgow Coma Score (GCS), platelet count, urine protein, and creatinine levels. Literature cited within the study emphasized the challenge of differentiating dengue from other febrile illnesses and the limitations of WHO classification systems in clinical settings. [1]

Ghosh et al. (2023) reviewed a range of epidemiological and predictive modeling literature, noting that while traditional time series models like ARIMA and its variants are widely used for infectious disease prediction, their accuracy improves with the addition of exogenous variables (e.g., rainfall). The authors build upon previous hybridization methods by combining ARIMAX with ARNN to model both linear and non-linear components of dengue incidence data. The model was tested on historical weekly dengue cases from San Juan and Iquitos, incorporating rainfall data as an exogenous input. It was compared against several benchmark models, including ARNNX, ETSX, TCNX, and TransformerX. [2]

The study by **Faridah et al. (2022)** conducted in Indonesia analyzed dynamic changes in platelet count over the course of dengue infection. Their retrospective analysis showed that declining platelet counts—particularly during the critical phase—correlated strongly with the development of dengue haemorrhagic fever. By identifying temporal trends in platelet dynamics, the authors highlighted the value of serial platelet monitoring for early detection of severe complications. This study underscores the potential of temporal data in enhancing clinical vigilance and preemptive care during dengue outbreaks. [3]

The study by **Tanner et al. (2008)** adopted a computational approach by applying decision tree algorithms to predict dengue diagnosis and outcomes in the early phase of illness. Using clinical data from patients in Vietnam, they trained and validated decision tree models capable of identifying key clinical features associated with confirmed dengue cases and severe outcomes. These features included platelet count, white blood cell count, tourniquet test results, and days of fever. The model demonstrated good predictive performance, supporting the use of machine learning techniques as an adjunct to clinical decision-making during dengue epidemics. [4]

Mayrose et al. (2023) proposed a machine learning framework using digital pathology to detect dengue from peripheral blood smear (PBS) images. The study utilized blob detection for platelet identification and various feature extraction techniques (morphological, GLSDM, LBP). Several classifiers were tested, achieving up to 95.74% accuracy with SVM. It stands out for its image-based ML approach, though limited by small sample size and exclusion of cytoplasmic features. [5]

Panja et al. (2022) introduced the XEWNet (Extended Wavelet Neural Network), combining wavelet decomposition (MODWT) with an ensemble neural network for dengue forecasting across San Juan, Iquitos, and Ahmedabad. Rainfall was found to be a causally significant predictor. This study surpassed traditional and deep learning models in both short-term and long-term forecasts. [6]

Tran & Xin (2024) evaluated transformer-based architectures for long-range dengue prediction using multivariate climate and case data from Singapore (2000–2019). The FWin Transformer (Fourier Mixed Window Attention) outperformed prior models (Autoformer, PatchTST, Fedformer) in both accuracy and forecasting duration. This paper offers a scalable architecture capable of handling non-stationary multivariate time series. [7]

He et al. (2024) addressed risk prediction in SFTS using a clinical dataset of 483 patients. Although focused on SFTS, the approach is directly applicable to dengue due to the clinical overlap. Their ML model (XGBoost) showed remarkable performance in predicting mortality, highlighting the role of early lab results and systemic symptoms in clinical prognosis. [8]

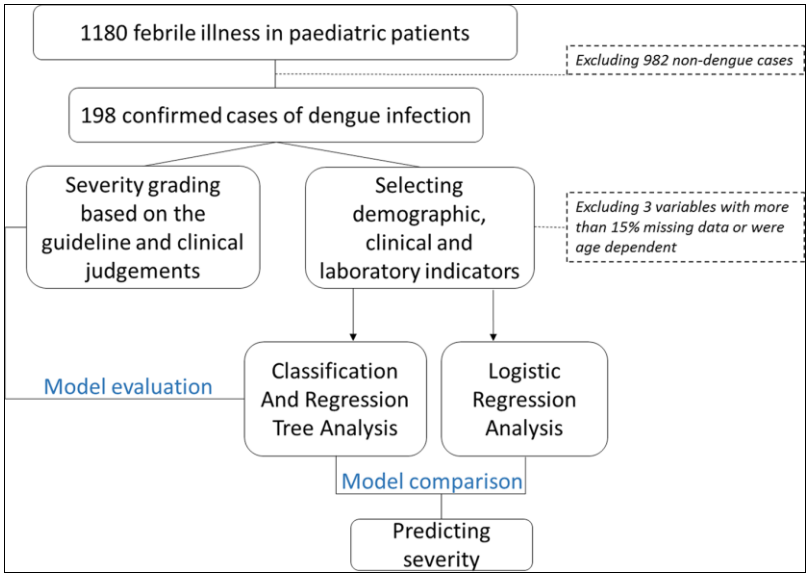
Ong et al. (2023) developed a dengue outbreak prediction model based on both vector indices and meteorological data across Kuala Lumpur districts. Their comparative analysis of seven ML classifiers found ensemble methods, especially XGBoost, to perform best. The Boruta algorithm was used to identify significant variables, revealing that the container index was the least important, improving the AUC by 6% upon removal. [9]

Silitonga et al. (2021) focused on predicting dengue severity levels (DF, DHF1, DHF2) using a small dataset of 77 patients from Indonesia. The study employed Artificial Neural Networks (ANN) and Discriminant Analysis (DA). ANN models using logistic and tanh activation functions achieved 90.91% accuracy, outperforming DA. This model aims to support early diagnosis during the third day of fever, which typically marks the transition to the critical phase. [10]

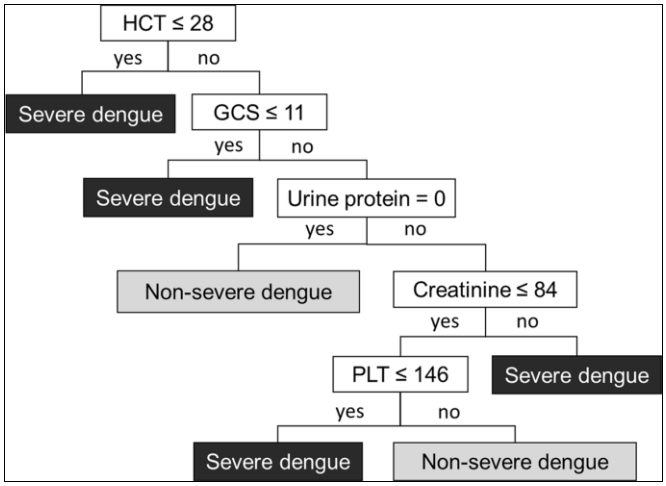
Hair et al. (2019) used unsupervised learning (SOM + Random Forest) to identify clinical patterns from 523 confirmed dengue cases in Brazil. Their clustering found four distinct groups correlating with disease severity, with age as the primary differentiator. Warning signs such as abdominal pain, hematocrit rise, and thrombocytopenia further differentiated cluster severity. The approach offers a novel method to uncover hidden clinical structures in dengue cases. [11]

2.3 Models / Algorithms used

The study by **Phakhounthong et al., 2018** contains a flowchart of the study as well as decision tree (CART model) diagram built using J48 algorithm.



(a) Flowchart of the study



(b) decision tree (CART model)

Fig 2.1: flowchart of the study and the decision tree (CART model)

Ghosh et al., 2023 contains a conceptual workflow of the hybrid ARIMA-ARNN model including data input, preprocessing, model layers, and output evaluation.

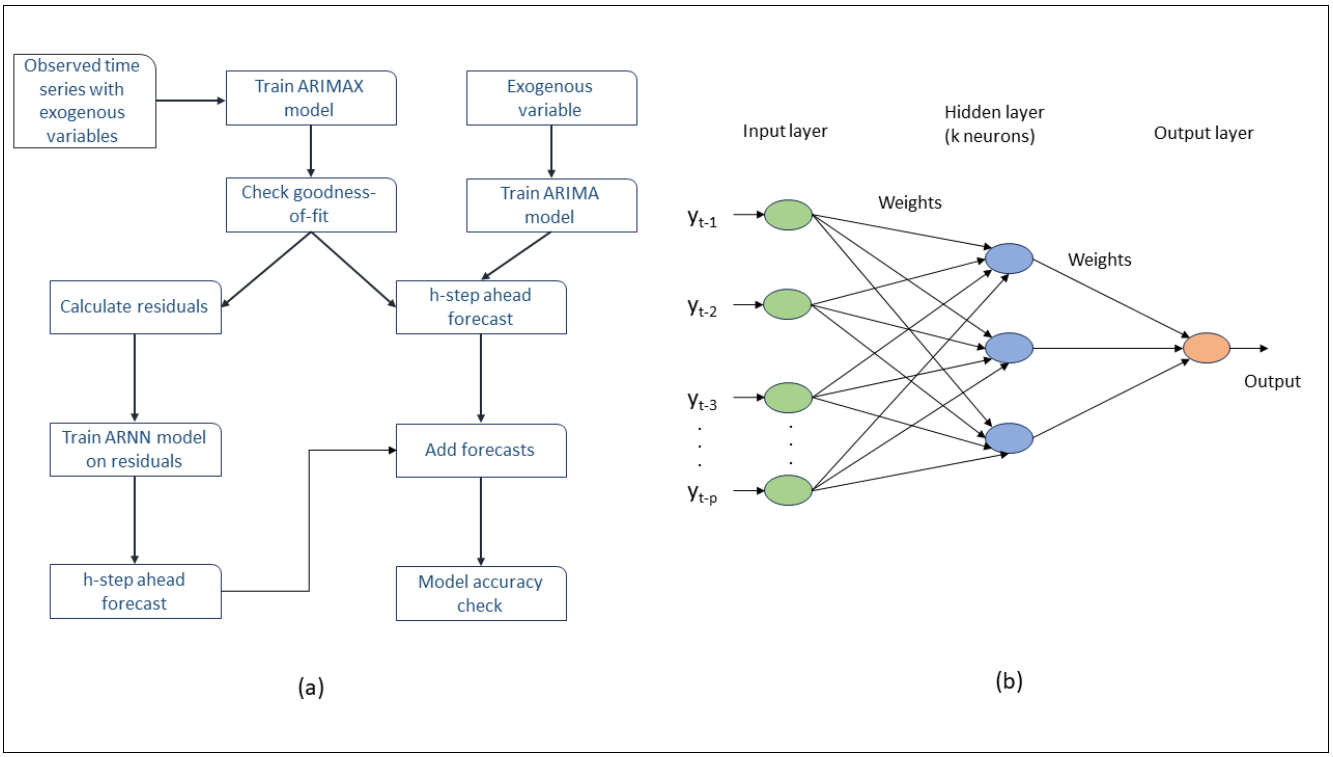


Fig 2.2: (a) Flow diagram of the proposed model (b) ARNN model structure.

The study by **Faridah et al. (2022)** contains a flowchart detailing the study's methodology and patient inclusion criteria is provided.

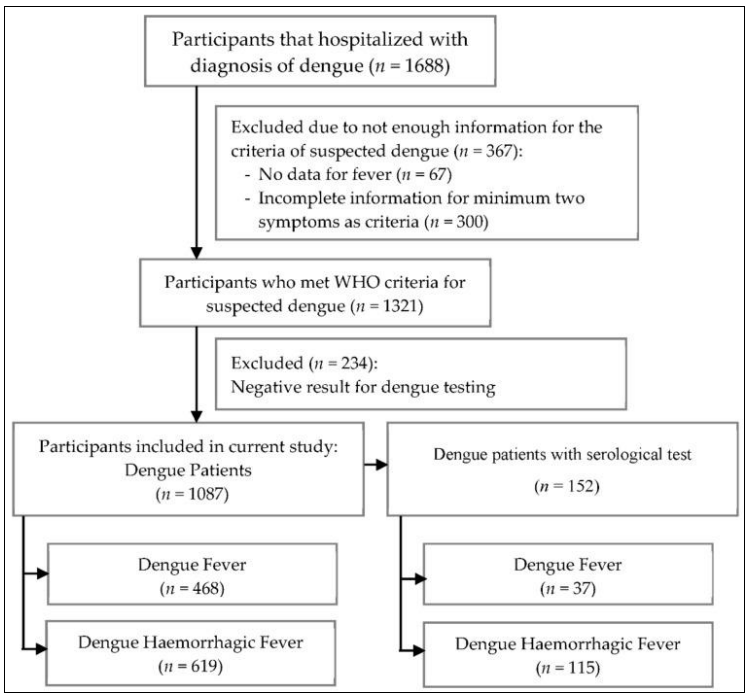


Fig 2.3: The flow chart for patients' selection and research design.

The study by **Tanner et al. (2008)** features a decision tree diagram representing the algorithm used for predicting dengue diagnosis and outcomes.

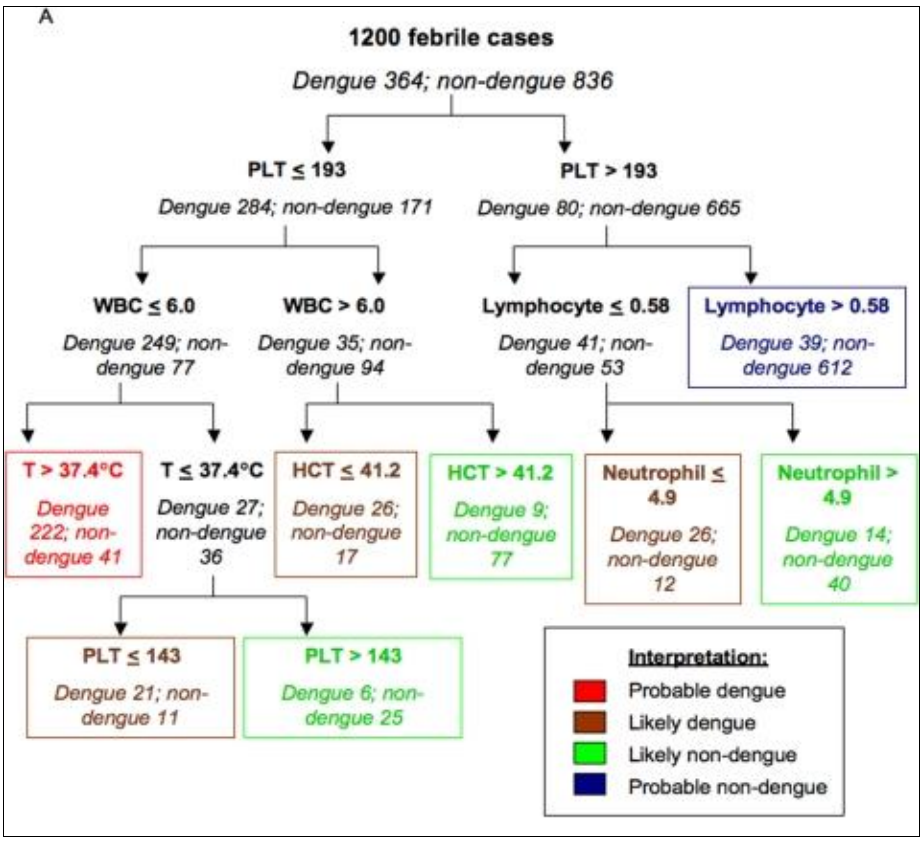


Fig 2.4: Decision algorithm for predicting dengue diagnosis

Mayrose et al., 2023 presents a processing pipeline for image-based dengue detection using platelet and lymphocyte features and ML classifiers

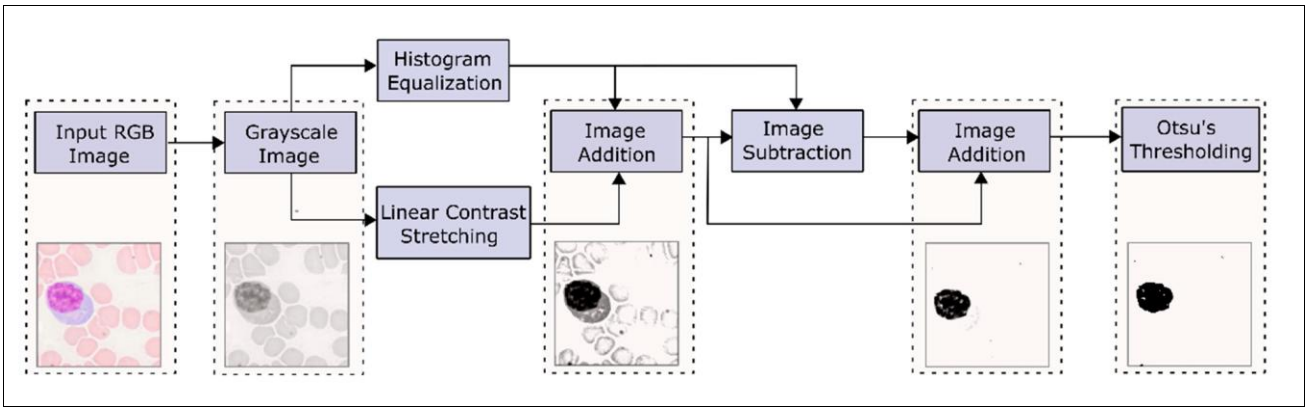


Fig 2.5: Processing pipeline for image-based dengue detection using platelet and lymphocyte features

Panja et al., 2022 includes a detailed model architecture diagram for the XEWNet model integrating wavelet transform, ANN, and exogenous climate variables

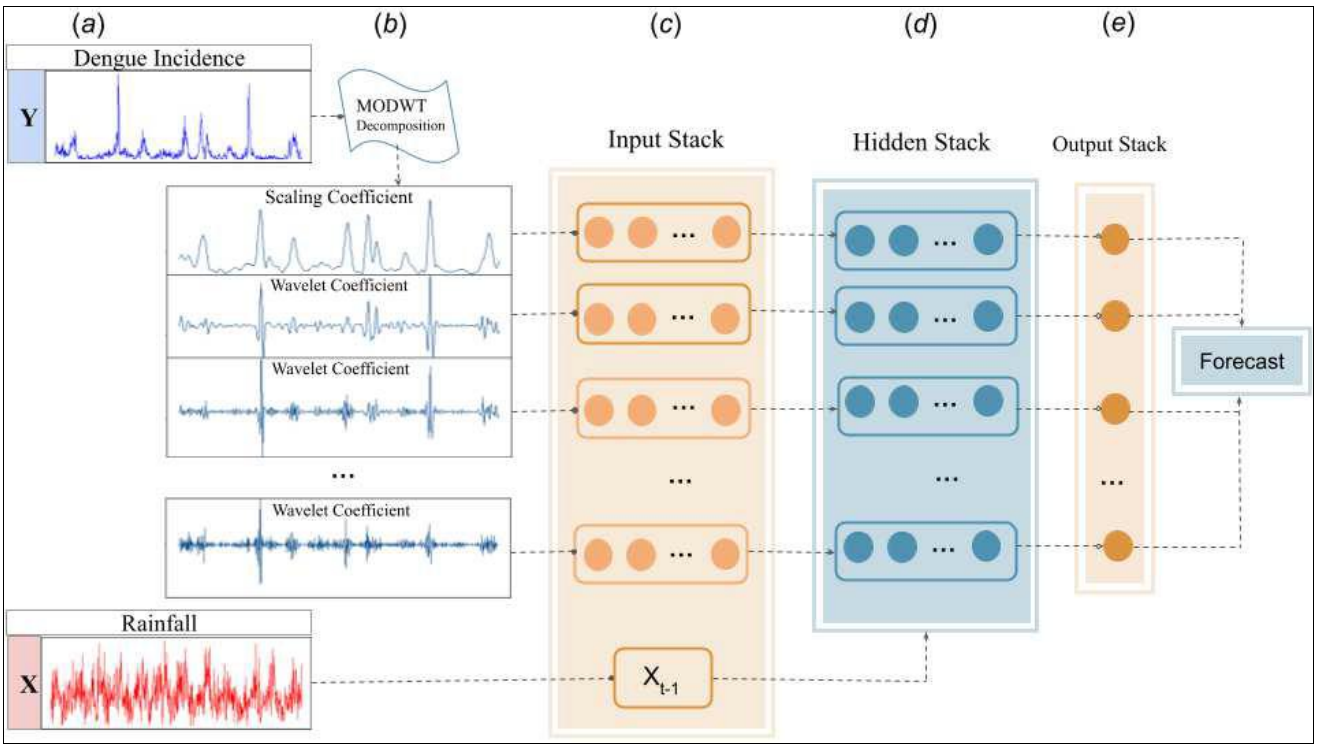


Fig 2.6: Model architecture diagram for the XEWNet model

Tran & Xin, 2024 presents the FWin model overview

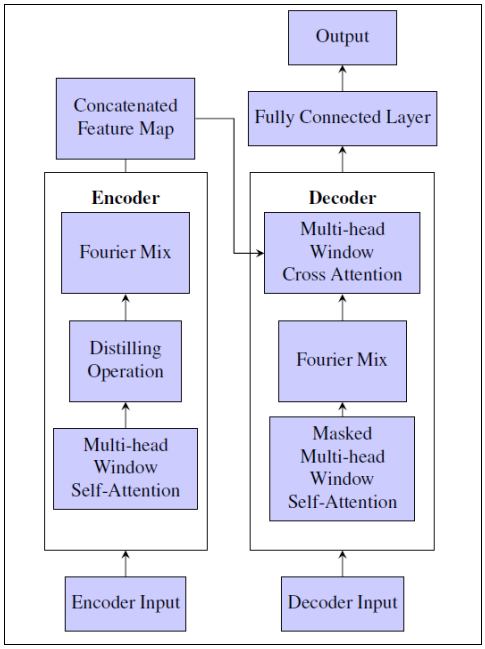


Fig 2.7: FWin model

The paper by **He et al. (2024)** contains a workflow diagram showing the data preprocessing, model building (six ML models), evaluation, and deployment of a web calculator.

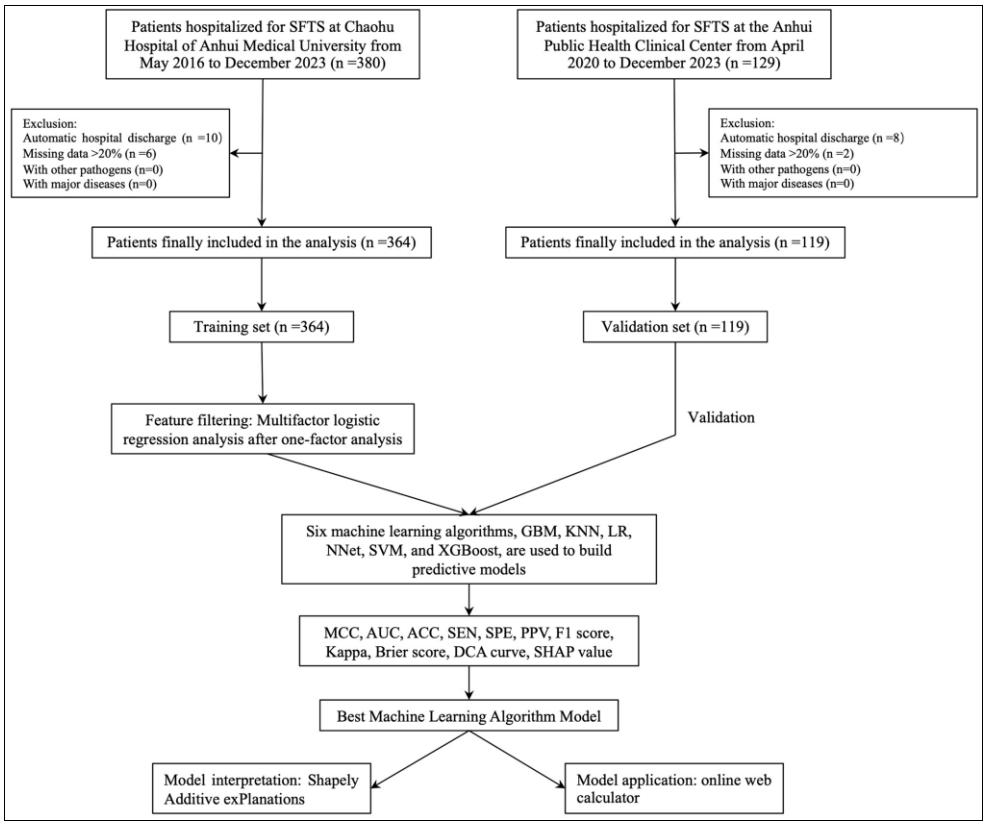


Fig 2.8: workflow diagram

The paper by **Ong et al., 2023** includes a workflow diagram for the development of machine learning classifiers using all variables or Boruta variables

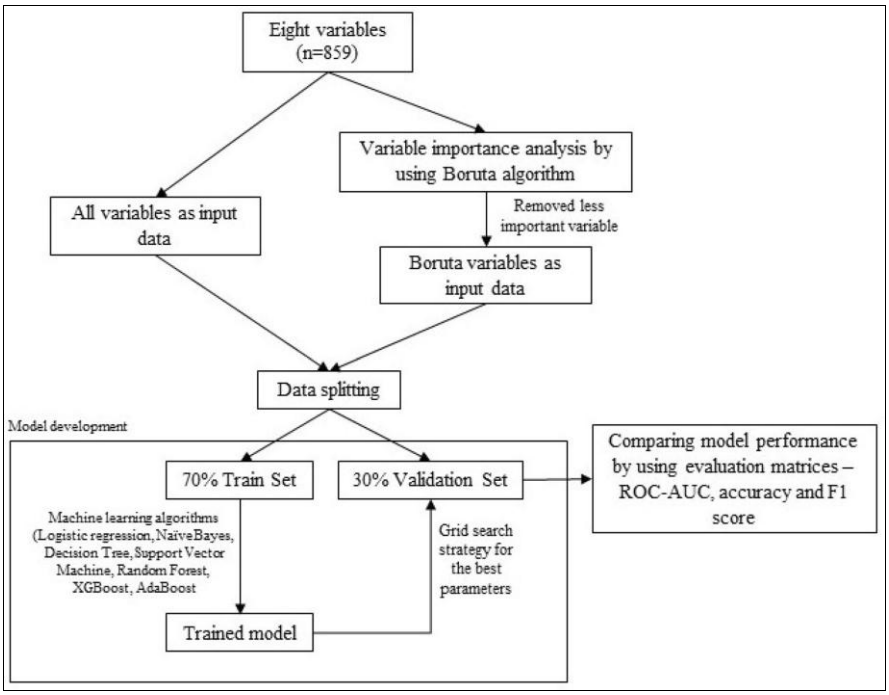
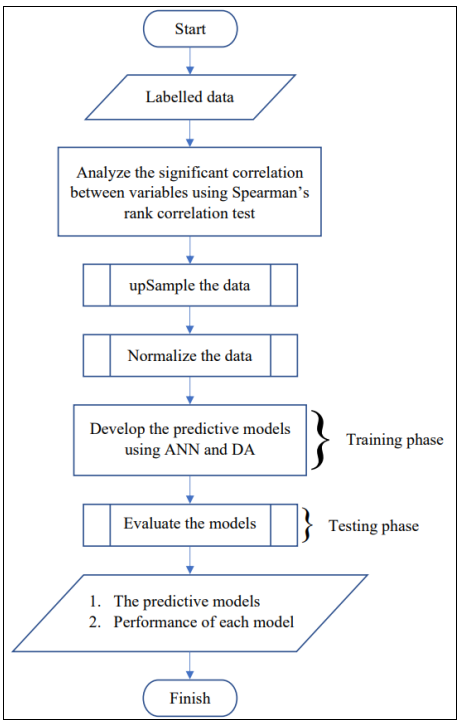
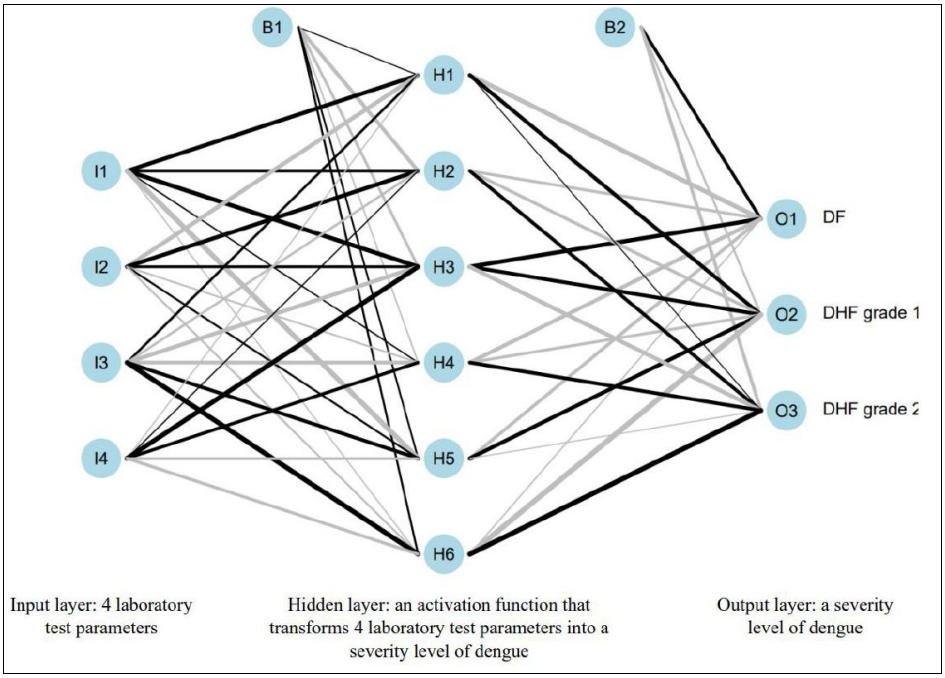


Fig 2.9: Workflow diagram

The paper by **Silitonga et al., 2021** features a model development flowchart & architecture of the predictive model.



(a) Model development flowchart



(b) ANN architecture

Fig 2.10: Model development flowchart and ANN architecture of the predictive model

The study by **Hair et al., 2019** Contains a workflow diagram for clinical dengue pattern classification

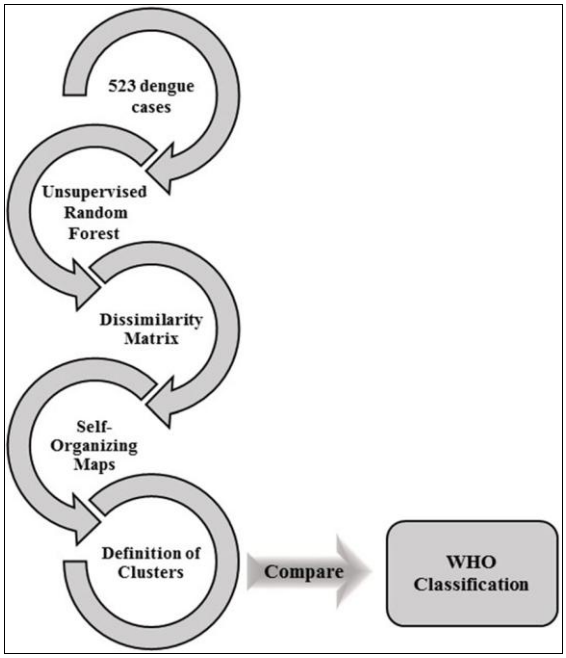


Fig 2.11: SOM + Random Forest model visualization and clustering

2.4 Findings & Gaps Identified

Paper Title	Key Findings	Gaps Identified
Predicting the severity of dengue fever in children on admission based on clinical features and laboratory indicators: application of classification tree analysis *(Phakhounthong et al., 2018)*	Developed a CART model using 5 variables (hematocrit, GCS, creatinine, urine protein, platelet count) with ~64% accuracy for predicting severe dengue.	Moderate accuracy; model limited to pediatric data and specific to Cambodian setting; not validated externally.
Anticipating dengue outbreaks using a novel hybrid ARIMA-ARNN model with exogenous variables *(Ghosh et al., 2023)*	Proposed hybrid model improved over ARIMA in many cases; maintained interpretability and robustness.	Limited to two regions (San Juan & Iquitos); did not explore spatial forecasting or uncertainty quantification.
Dynamic Changes of Platelet and Factors Related Dengue Haemorrhagic Fever: A Retrospective Study in Indonesian *Faridah et al. (2022)*	Identified dynamic changes in platelet counts and other hematological parameters associated with progression to dengue hemorrhagic fever.	Retrospective nature limits real-time applicability; requires prospective studies for validation.
Decision tree algorithms predict the diagnosis and outcome of dengue fever in the early phase of illness *Tanner et al. (2008)*	Developed decision tree models using clinical and hematological parameters to predict dengue diagnosis and outcomes with reasonable accuracy.	Early-phase data may not capture full disease progression; models need validation in diverse populations.
Machine learning-based detection of dengue from blood smear images utilizing platelet and lymphocyte characteristics	SVM and Decision Tree achieved 93.6% accuracy using lymphocyte/platelet features	Small dataset; requires external dataset validation; limited generalizability to

(Mayrose et al., 2023)	from blood smear images.	other image sources.
An ensemble neural network approach to forecast dengue outbreak based on climatic condition *(Panja et al., 2022)*	Outperformed other statistical and ML methods in 75% of scenarios across 3 cities using rainfall data.	Only one exogenous factor (rainfall) used; model complexity may hinder real-world deployment.
FWin Transformer for Dengue Forecasting Under Climate & Ocean Influence *(Tran & Xin, 2024)*	Proposed FWin Transformer showed superior long-range dengue forecasting up to 60 weeks.	Does not model spatial interactions; requires further validation in other regions beyond Singapore.
Machine Learning for Identifying Risk of Death in Patients with severe fever with thrombocytopenia syndrome *(He et al., 2024)*	Developed ML models (XGBoost, LightGBM) for death risk prediction; ensemble model achieved >90% AUC.	Focused on SFTS, not dengue directly; generalization to other diseases (like dengue) not demonstrated.
Predicting dengue transmission rates by comparing different machine learning models with vector indices and meteorological data *(Ong et al., 2023)*	Ensemble models (XGBoost, RF, AdaBoost) outperformed classical ML models; removing less important variables improved performance.	Did not incorporate spatial or temporal autocorrelation in model; lacked socioeconomic or mobility data.
Comparison of dengue predictive models developed using artificial neural network and discriminant analysis with small dataset *(Silitonga et al., 2021)*	ANN (logistic and tanh activation) achieved 91% sensitivity and 95% specificity using small clinical dataset.	Limited sample size; lacks external validation; no inclusion of time series or external covariates.

Characterization of clinical patterns of dengue patients using an unsupervised machine learning approach *(Hair et al., 2019)*	SOM + Random Forest clustered dengue cases into 4 groups by severity; age was major determinant.	Cross-sectional only; no time-based disease progression modeling; requires longitudinal validation.

Table 2.1: Key finding and Gap Identified based on the Literature Review

Chapter 3: Methodology

3.1 Introduction

This chapter gives an outline of methods that were followed during the project to attain the goal. It included data sources & format, data preprocessing techniques, data exploration and analysis, process description, workflow and requirements of hardware & software for completion of the project. It provides information about the dataset and from where it was taken. Workflow of the project which gives a brief understanding about what exactly is done in the project. Data visualization is done to understand how data appears.

3.2 Data Sources & Format

The dataset used in this study was obtained from **Fr. Muller's Medical College Hospital**, Mangalore, as part of a clinically supervised dengue study. The data collection was conducted in accordance with ethical research guidelines and was approved by the institutional ethics committee under **Protocol No: FMMCIEC/CCM/149/2019**. The dataset comprises clinical records of **305 dengue-positive patients**, and includes both demographic details and time-series medical test results. The dataset was shared in **CSV format** and permission was obtained prior to its use for research and analysis purposes, ensuring compliance with data privacy and ethical standards. The dataset includes the following types of variables:

Variable	**Description**	**Type**
p_no.	Unique patient identifier	Numerical
Age (Years)	Age of the patient in years	Numerical
Sex	Biological sex of the patient (Male/Female)	Categorical
Date of Onset	Date when the patient first showed symptoms	Object
Date of Discharge	Date when the patient was discharged	Object

Symptoms	Text field listing patient symptoms	Object
Date of Test & Time of Test	Date and time when a test was conducted	Object
Haemoglobin (gm/Dl)	Haemoglobin level (g/dL)	Object
Platelet (cells/cu.mm)	Platelet count (cells/µL)	Object
Red Blood Cell Count (millions/cu.mm)	Red Blood Cell count	Object
Haematocrit (Packed Cell Volume) (%)	Percentage of red blood cells in blood	Object

Table 3.1: Description of the dataset

All personally identifiable information (PII) was removed prior to analysis to maintain patient confidentiality. The data was used solely for academic and research purposes with strict adherence to data privacy and ethical standards.

3.2.1 Acknowledgement

The author sincerely acknowledges the support and guidance provided by **Fr. Muller's Medical College Hospital** in granting access to anonymized patient records for this study. Special thanks to the hospital's Ethics Committee for providing ethical clearance under **Protocol No: FMMCIEC/CCM/149/2019**, which enabled the responsible use of clinical data for research. Gratitude is also extended to the medical staff and data coordinators who assisted in data organization and validation.

3.3 Data Preprocessing – Extraction & preparation

In order to prepare the raw dataset of dengue patients for predictive modelling, a series of data preprocessing steps were conducted to transform the unstructured information into a clean,

analysable format. Initially, the dataset was standardized by renaming columns for consistency and forward-filling the missing values in the patient identification column.

Key lab test columns such as *Haematocrit*, *RBC*, *Platelet*, and *Haemoglobin* were examined for non-numeric entries like “not done”, which were replaced with NaN values, followed by converting these columns into appropriate numerical data types. The *Sex* column was also cleaned to ensure binary categorization (i.e., only "Male" and "Female"), and all date and time entries were validated, cleaned, and then transformed into separate *Date* and *Time* columns using the datetime format.

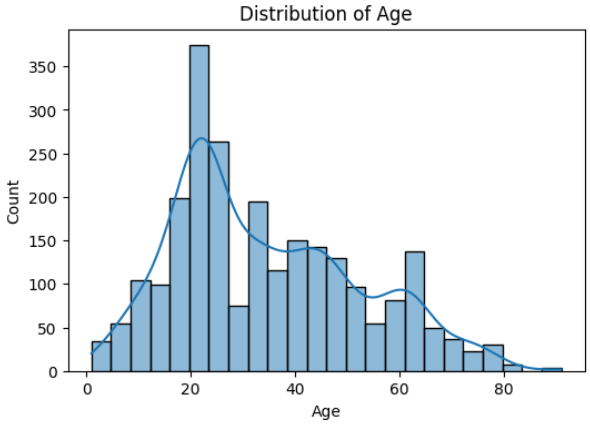
Further refinement included ensuring each patient had a unique identifier. In instances where the same patient number was assigned to multiple records with different age values, the records were split and reindexed so that each patient had a distinct ID. The *Symptoms* column, which contained free-text entries, was processed by extracting unique symptom descriptors for each patient, standardizing their format (e.g., capitalizing the first letter), and transforming them into individual binary columns, thereby creating a structured feature set.

Additional features were engineered, such as *days_since_onset*, to quantify disease progression, and lag features for *Platelet* and *Haematocrit* values to help model temporal trends. Finally, categorical features like *Sex* were encoded numerically to prepare the data for machine learning algorithms. These preprocessing steps significantly improved the quality and utility of the dataset, enabling more accurate and reliable predictions.

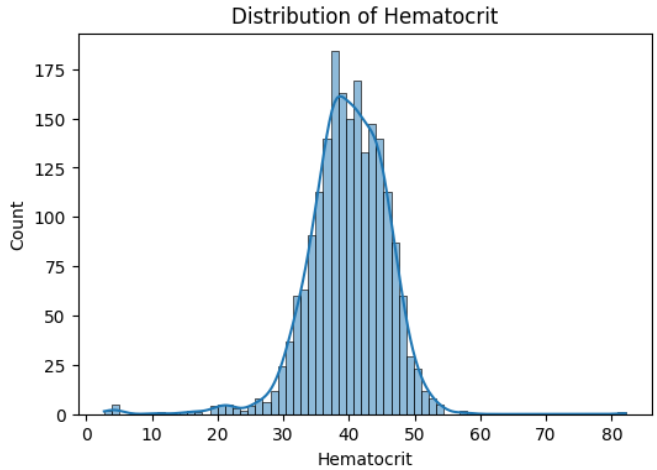
3.4 Data Exploration & analysis

Data exploration and analysis form a critical part of understanding trends, distributions, and patterns in any dataset. In this study, we performed a comprehensive analysis of clinical and demographic features associated with dengue patients, focusing particularly on platelet count trends. The visualizations generated during the analysis provide valuable insights, summarized under the following key areas:

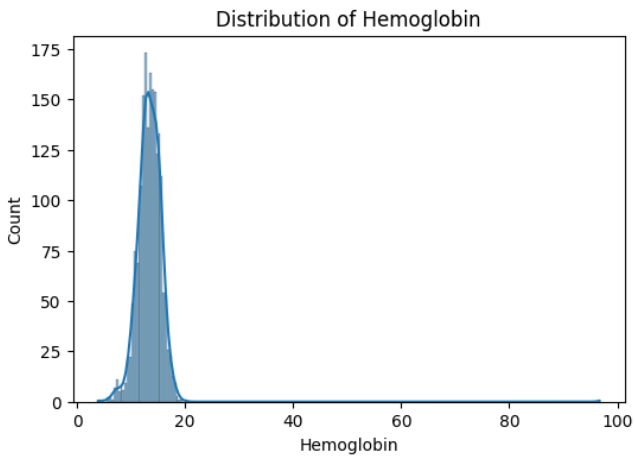
3.4.1 Univariate Analysis



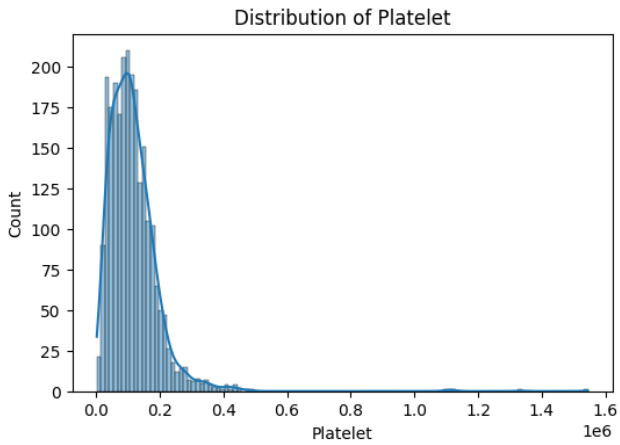
(a) Age distribution



(b) Hematocirt distribution



(c) Hemoglobin distribution



(d) Platelet distribution

Fig 12: Univarite Analysis

Univariate analysis was performed to understand the distribution of key individual lab features in the dataset. The histograms and density plots offer insights into the central tendency, spread, and skewness of the variables:

- Age: The distribution of age is right-skewed, with a high concentration of patients in the younger age group (10–30 years), reflecting the known vulnerability of adolescents and young adults to dengue infection.
- Hematocrit: The distribution appears approximately normal, centered around 40%. This is within expected physiological ranges, although outliers at both ends may indicate hematological complications in some cases.
- Hemoglobin: The values cluster tightly around normal reference ranges (10–18 g/dL), though a few high values may suggest data entry errors or rare clinical cases.
- Platelet Count: This variable is highly right-skewed, with most values concentrated below 250,000/µL. A few extreme outliers exceed 1,000,000/µL, which could be abnormal physiological responses or data quality issues.

This univariate exploration helped identify the presence of skewed distributions, potential outliers, and the need for transformation or scaling during preprocessing.

3.4.2 Time Based Analysis

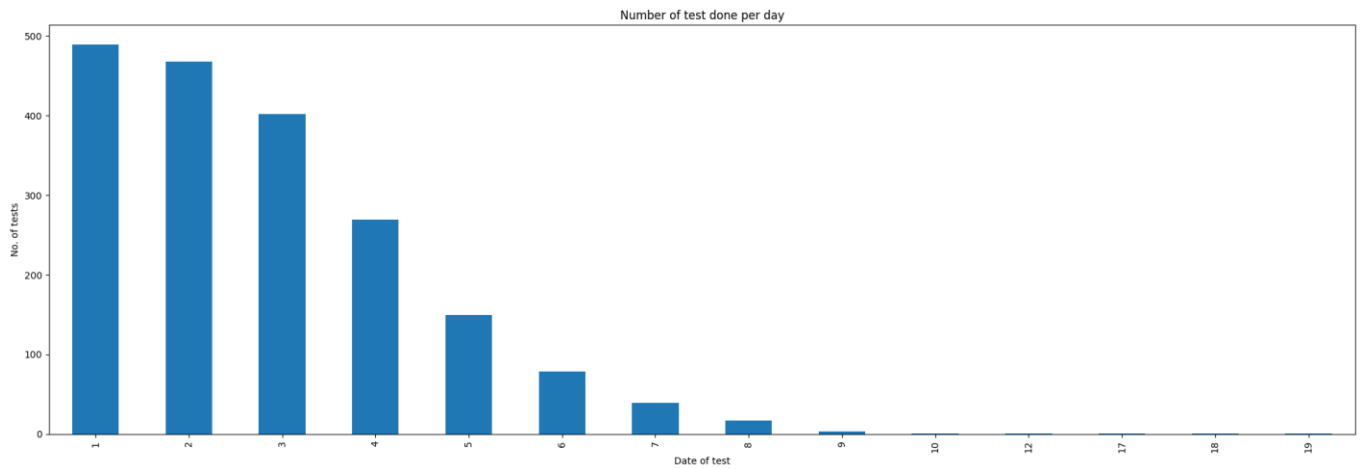


Fig 13: Number of test done per day

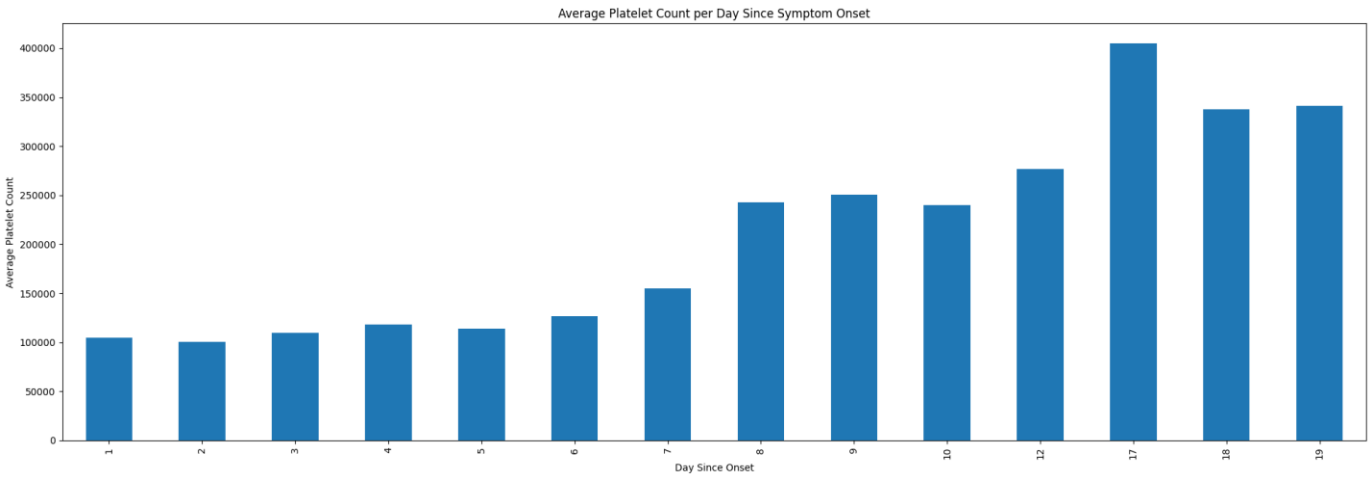


Fig 14: Average platelet count per day since onset

Time-based analysis was conducted to evaluate the temporal patterns in test frequency and platelet count progression relative to the onset of dengue symptoms.

- Number of Tests per Day: The upper plot reveals that the majority of diagnostic tests were conducted within the first 3–5 days after symptom onset, peaking around Day 1. This aligns with clinical practice, as early testing is crucial for confirming dengue diagnosis and initiating timely management.
- Average Platelet Count per Day Since Onset: The lower plot shows that average platelet counts remain relatively stable in the early days but begin to rise significantly after Day 6. This trend indicates the typical recovery pattern in dengue patients, as platelet levels drop during the critical phase (Day 3–6) and begin to improve afterward.

This temporal analysis confirms both clinical expectations and data reliability. It also highlights the importance of monitoring platelet trends closely between Days 3 to 7 when the risk of complications is highest.

3.4.3 Age and Gender Demographics

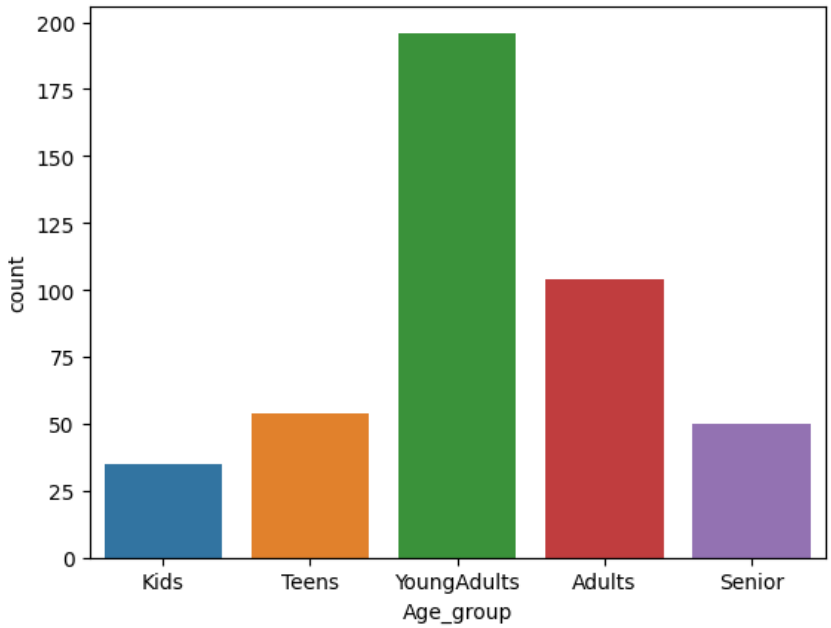


Fig 15: Age group distribution

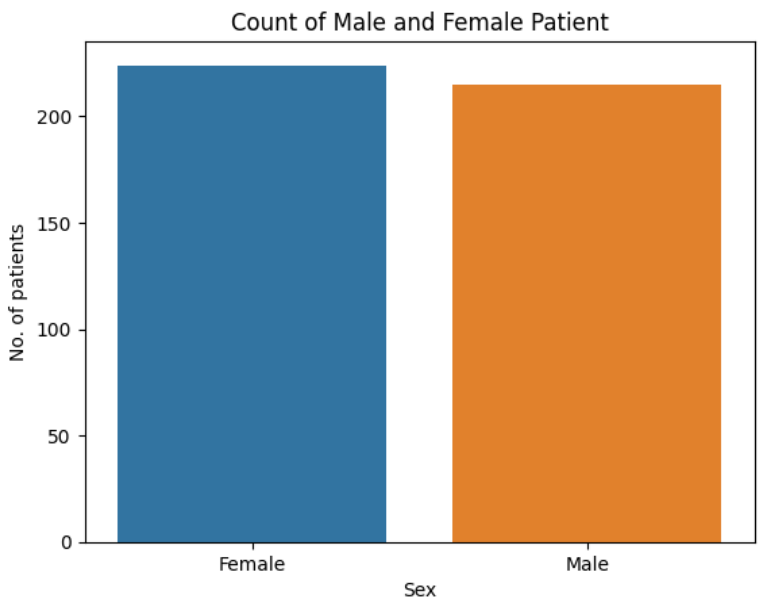


Fig 16: Count of Male and Female Patients

Understanding the demographic distribution of dengue patients helps identify the most affected groups and tailor public health interventions accordingly.

- Age Group Distribution: As shown in the left plot, the majority of dengue-positive cases fall under the Young Adult category, followed by Adults and Seniors. Teens and Kids comprise a smaller portion of the affected population. This suggests that individuals aged between 20–40 years are the most vulnerable or most frequently tested during dengue outbreaks.
- Gender Distribution: The right plot depicts a nearly balanced gender ratio, with a slightly higher number of female patients compared to male patients. This indicates a uniform distribution of dengue cases across sexes, with no major gender bias in test frequency or positivity.

This demographic insight aids in risk stratification and emphasizes the need for awareness campaigns targeted toward young adults and working-age individuals.

3.4.4 Symptom Based Analysis

This section focuses on understanding how various symptom categories are distributed across patients, age groups, and genders, as well as how they co-occur.

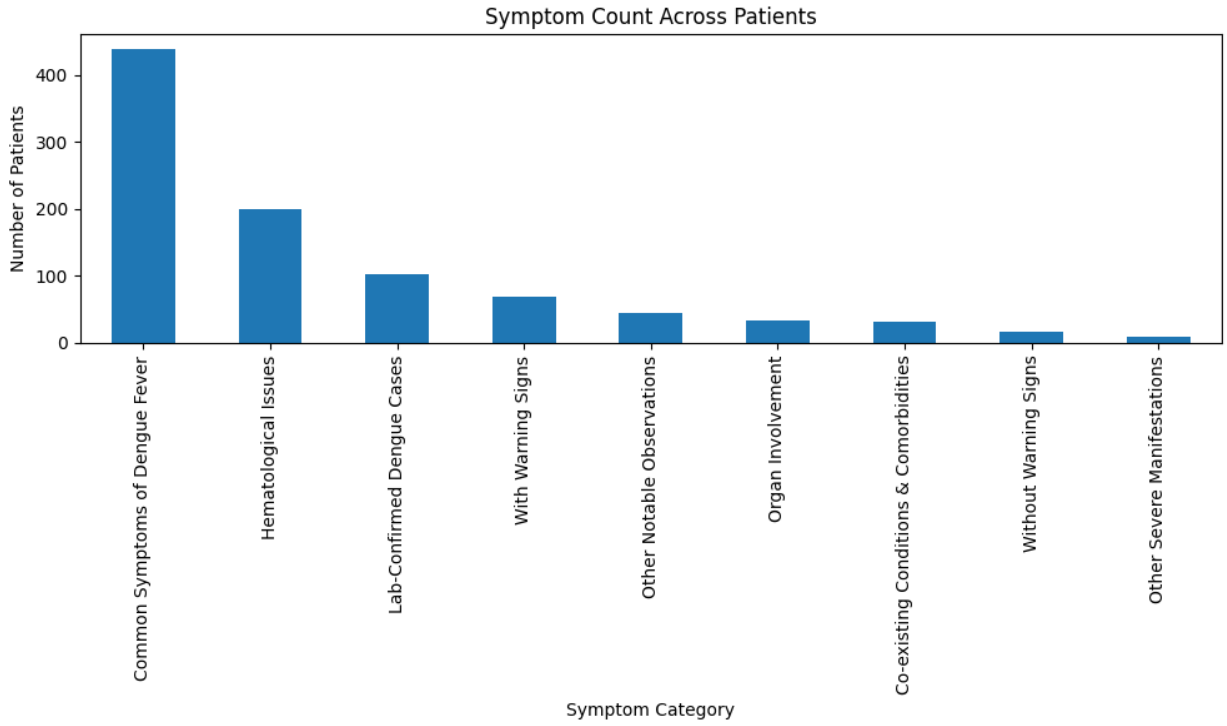


Fig 17: Symptoms count across patients

Symptom Distribution Across Patients

The bar plot shows the total number of patients falling under each major symptom category:

- The majority of patients presented with Common Symptoms of Dengue Fever such as fever, headache, and body aches.
- Hematological issues (e.g., thrombocytopenia and leukopenia) and Lab-confirmed dengue cases also formed a significant portion.
- Symptoms with warning signs, organ involvement, and comorbidities were less frequent but clinically important, indicating progression or severity.

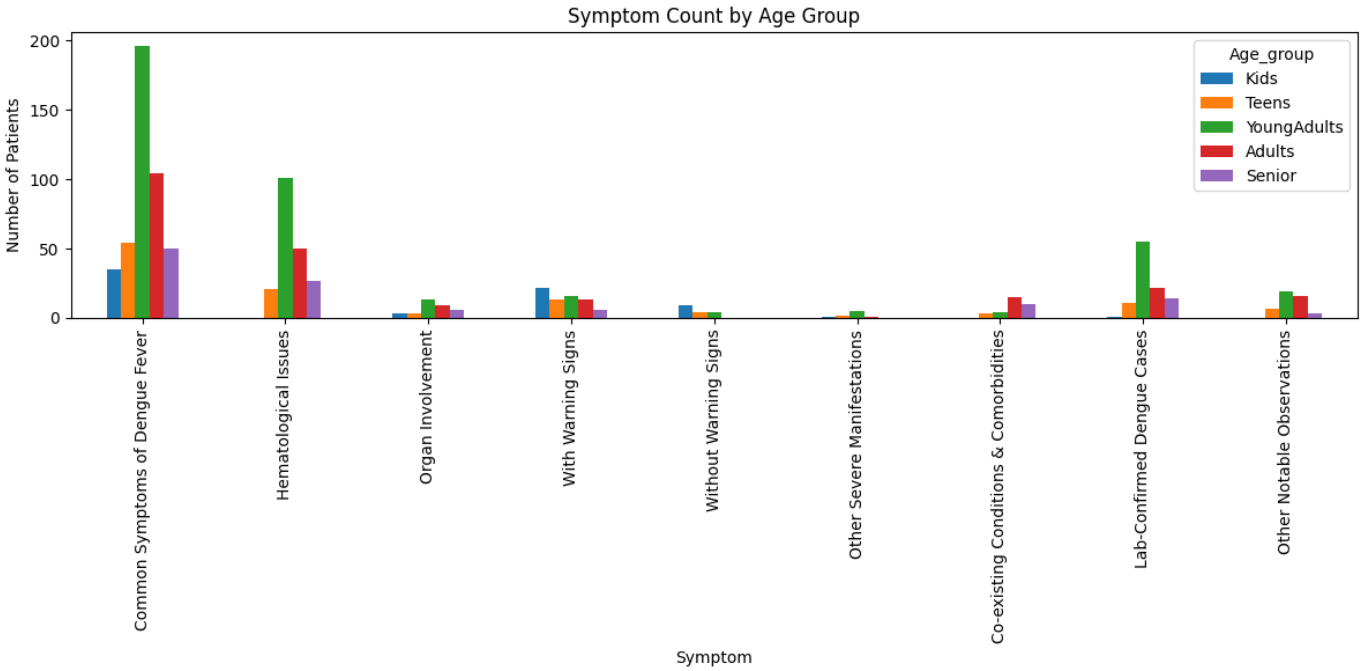


Fig 18: Symptoms count by age group

Symptom Count by Age Group

When symptoms were grouped by age:

- Young Adults were the most affected across all categories, particularly for common and hematological symptoms.
- Children (Kids and Teens) had fewer complications like organ involvement or comorbidities, while Seniors showed relatively higher rates of co-existing conditions.

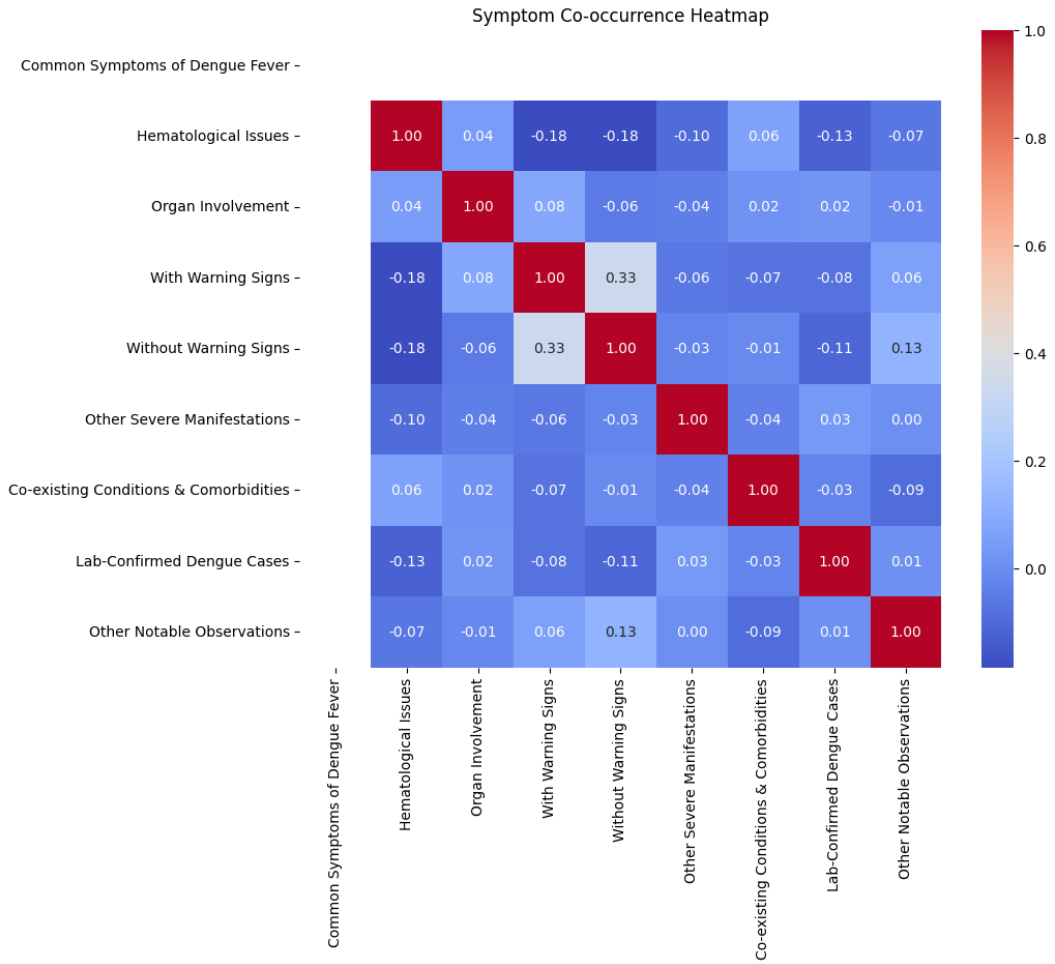


Fig 19: Symptoms co-occurrence heatmap

Symptom Co-occurrence Heatmap

The co-occurrence matrix helps understand how often different symptom categories appear together:

- Hematological Issues and Warning Signs had mild positive correlation.
- With and Without Warning Signs showed a moderate inverse relationship.
- The absence of strong correlations across most categories suggests that symptom manifestation varies greatly by individual case.

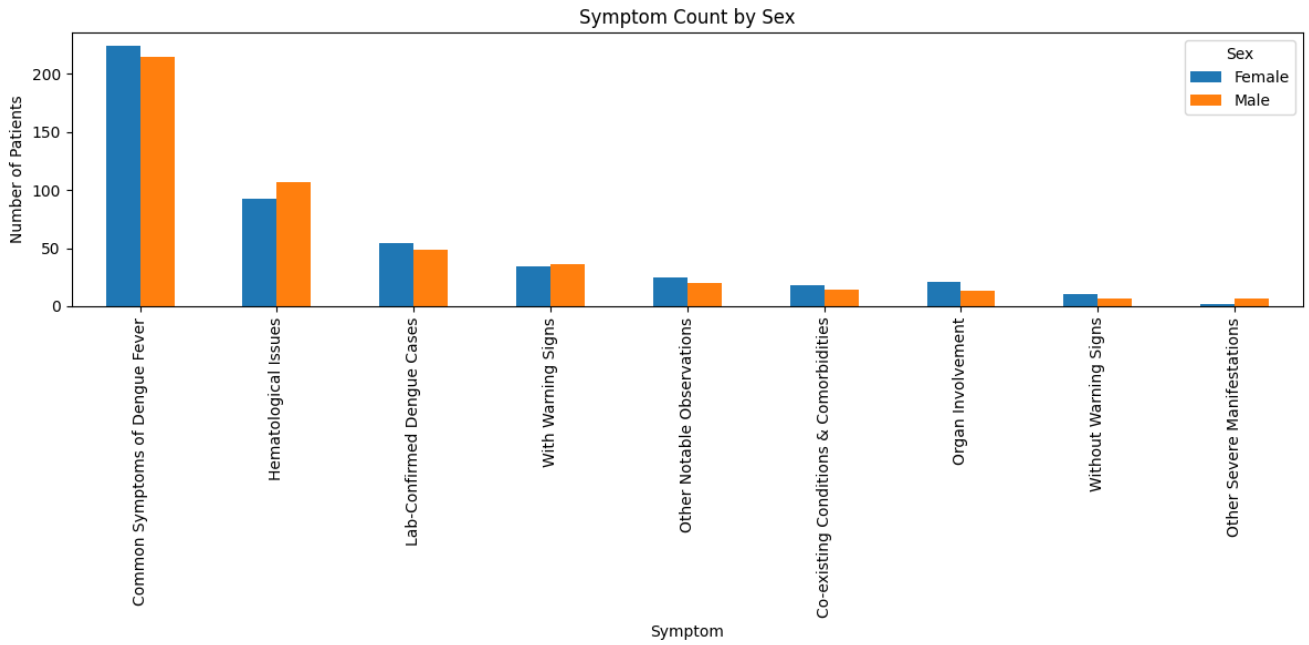


Fig 20: Symptoms count by sex

Symptom Count by Sex

Sex-wise analysis showed:

- A fairly balanced distribution between female and male patients across all symptom categories.
- Slightly higher counts in females for common symptoms and hematological issues, possibly due to sample size or healthcare-seeking behavior.

3.4.5 Lab Test Feature Distributions

This section explores the distribution and interrelationships of key lab parameters—platelet count, hematocrit, hemoglobin, and RBC—which are essential clinical markers for tracking dengue progression.

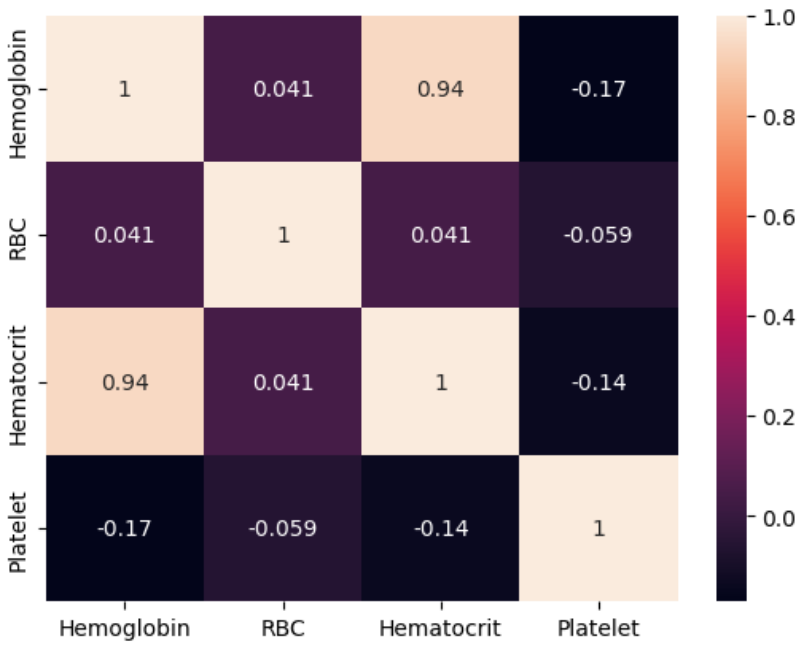


Fig 21: Correlation heatmap of the lab test

Correlation Heatmap

The correlation matrix highlights the following:

- A very strong positive correlation (r = 0.94) between hemoglobin and hematocrit, confirming biological consistency, as both derive from red blood cell volume.
- Weak or no correlation between platelet count and the other lab variables, indicating that platelet levels fluctuate independently during disease progression. This justifies treating platelet count as a standalone target variable in prediction tasks.
- Minimal correlation between RBC and other parameters, suggesting relatively stable RBC levels across patients.

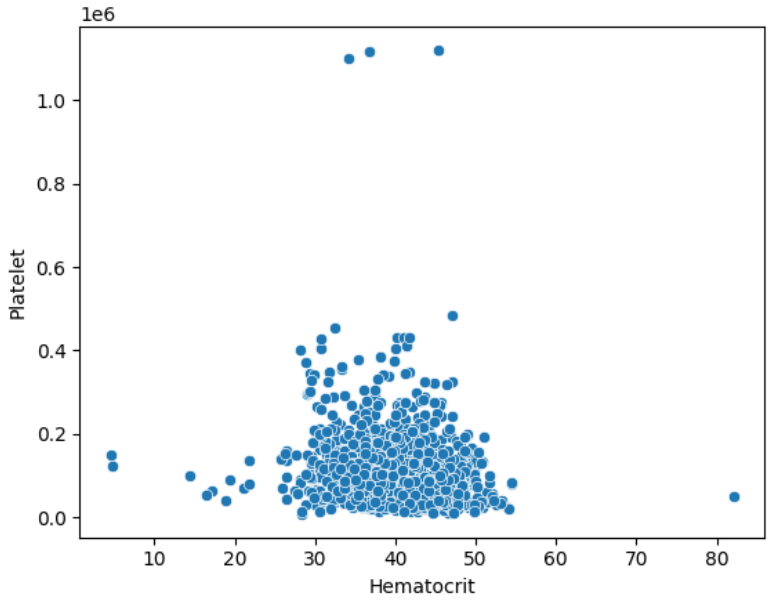
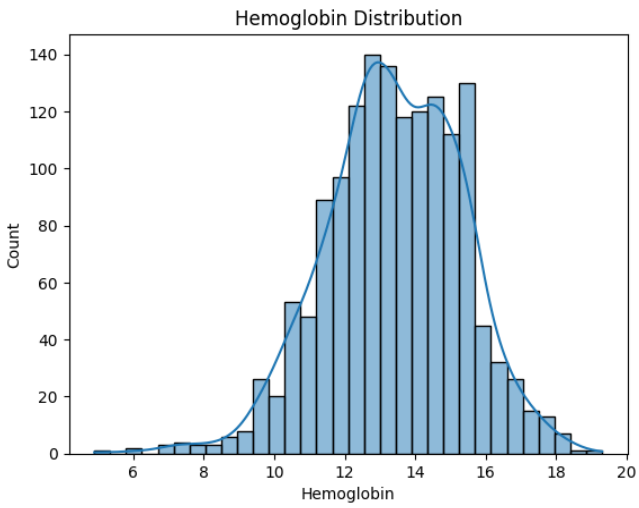


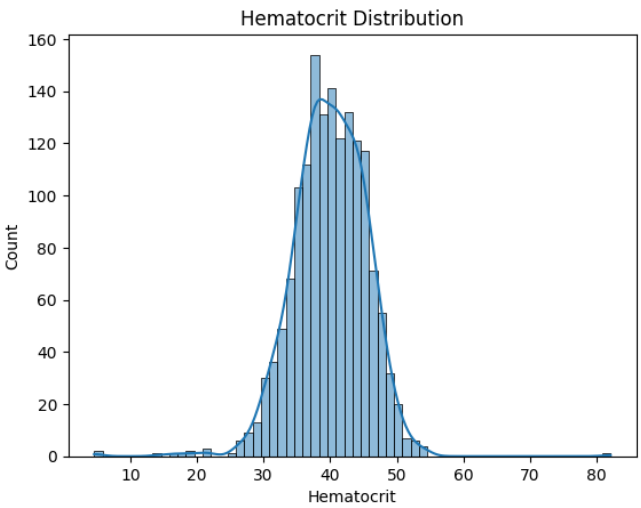
Fig 22: Scatter plot of platelet and hematocrit

Scatter Plot – Platelet vs. Hematocrit

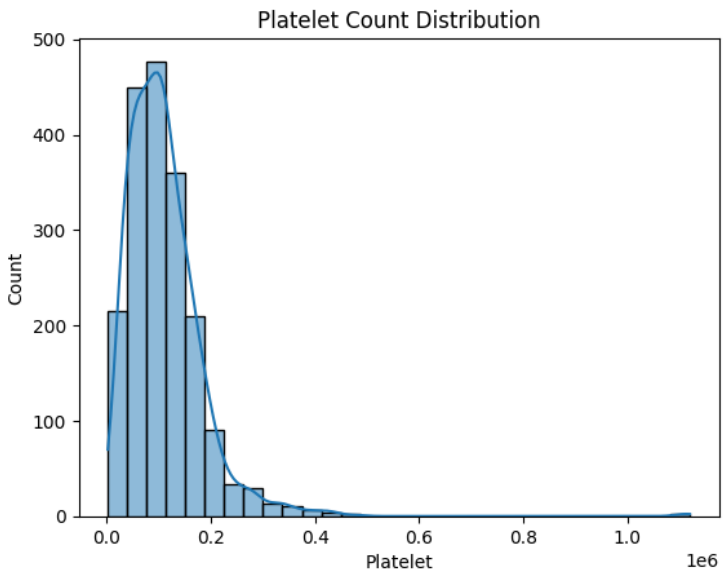
The scatter plot provides insight into the interaction between platelet count and hematocrit levels. While there is no obvious linear correlation, a concentration of points with low platelet counts and moderate to high hematocrit suggests plasma leakage and thrombocytopenia occurring together—a hallmark of severe dengue cases. This relationship, though weak, can aid in rule-based clinical classifications.



(a) Hemoglobin count distribution



(b) Hematocrit count distribution



(c) Platelet count distribution

Fig 23: Lab test distributions

Distributions:

- Hematocrit Distribution: The histogram shows a normal-like distribution, with most values concentrated between 35% and 45%. This suggests that the majority of patients had hematocrit levels within or near the normal range, though extreme values could indicate dehydration or internal bleeding.
- Hemoglobin Distribution: Hemoglobin values also follow a bell-shaped curve, ranging mostly between 10–16 g/dL. This supports the reliability of data and reflects typical clinical expectations, with a few low values possibly indicating anemia.
- Platelet Count Distribution: Platelet count is highly right-skewed, with most patients falling below 250,000/µL. A few extremely high values may be due to lab anomalies or input errors. This skewness justifies the need for transformation or normalization before modeling.

3.5 Process Description

This section elaborates the entire process involved in the development and functioning of the platelet count forecasting system for dengue patients. The proposed software system leverages

historical patient data, symptoms, and laboratory values to predict future platelet counts using data-driven forecasting models. The software is modular in nature and each module is responsible for a specific task in the data pipeline, from data ingestion to prediction visualization. Below is a detailed breakdown of the modules and the workflow logic.

Overview of the System

The platelet forecasting system comprises the following key modules:

1. Data Collection and Preprocessing Module
2. Feature Engineering Module
3. Exploratory Data Analysis (EDA) Module
4. Model Development and Training Module
5. Platelet Forecasting Module
6. Visualization and Output Interface

Each module interacts with others in a linear or feedback loop as necessary, forming a coherent data processing pipeline. The system is designed to handle tabular datasets containing demographic, temporal, clinical, and laboratory test features for dengue patients.

Module Descriptions and Logic

1. Data Collection and Preprocessing Module

Function:
This module is responsible for loading and validating the dataset, handling missing or inconsistent data, and converting it into a machine-readable format.

Inputs: Raw CSV/XLSX dataset containing:

- Patient ID, age, sex
- Date of admission/discharge, date and time of test
- Clinical symptoms (fever, rash, bleeding, etc.)
- Lab test values (platelet count, haemoglobin, RBC, haematocrit)

Process Logic:

- Convert date fields to datetime objects.
- Normalize text fields (e.g., symptom descriptions).
- Impute missing values using statistical or regression-based methods.
- Group patient data by ID and order by timestamp to maintain temporal continuity.

2. Feature Engineering Module

Function:

Enhances raw data by creating derived features that can improve model performance.

Process Logic:

- Generate time-lagged features for platelet count (e.g., previous day counts).
- Create duration features (e.g., number of days since symptom onset).
- Categorize age into brackets (e.g., child, adult, elderly).
- Encode sex as binary.
- Aggregate symptom combinations to generate severity scores.
- Normalize lab values using min-max or z-score normalization.

3. Exploratory Data Analysis (EDA) Module

Function:

Performs statistical and visual analysis to understand patterns and guide model development.

Process Logic:

- Visualize platelet trends over time for individual patients.
- Correlation heatmaps to understand relationships between features.
- Identify outliers using boxplots and statistical thresholds.
- Analyze symptom frequency and co-occurrence.

4. Model Development and Training Module

Function:

Builds and trains machine learning models to forecast platelet counts.

Process Logic:

- Split dataset into training and testing sets (with patient-wise split to avoid data leakage).
- Use time-series forecasting models (e.g., LSTM, Random Forest Regressor, or XGBoost).
- Train models on sequential data with features selected from engineering step.
- Perform hyperparameter tuning using cross-validation techniques.
- Evaluate models using RMSE, MAE, and R^2 scores.

5. Platelet Forecasting Module

Function:

Generates predictions for future platelet counts based on the trained model.

Process Logic:

- For a given patient, fetch the latest temporal data and feed it into the model.
- Predict platelet counts for the next 1–3 days.
- Compare predicted vs. actual (if available) to monitor accuracy.
- Optionally, provide risk stratification (e.g., "critical", "low-risk") based on predicted values.

6. Visualization and Output Interface

Function:

Displays the prediction results and other key insights in a user-friendly format.

Process Logic:

- Plot predicted vs. actual platelet count trends.
- Interactive charts for lab value trends and symptom timelines.

3.6 Workflow diagram

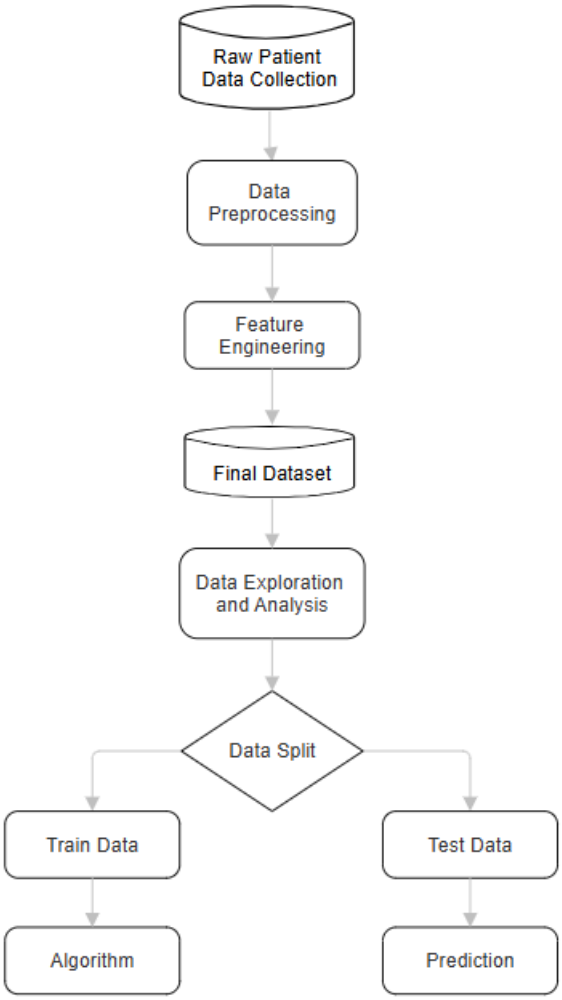


Fig 24: Workflow diagram

3.7 Hardware & Software specifications

This section outlines the complete hardware and software stack used for the development and deployment of the platelet count forecasting system for dengue patients. It includes the programming languages, libraries, frameworks, and data processing tools employed, as well as variable definitions and hardware configurations used during development and testing.

Software Specifications

Programming Languages Used

- **Python 3.10**: The primary programming language used for data preprocessing, feature engineering, model development, and prediction. Chosen for its simplicity, rich ecosystem, and extensive data science support.

Python Packages and Libraries

Package	Purpose/Usage
pandas	For data manipulation and analysis (cleaning, transformation, grouping).
numpy	For numerical operations and efficient array handling.
matplotlib / seaborn	For data visualization and plotting platelet trends.
scikit-learn	For machine learning model development (Random Forest, Regression, etc.).
xgboost	Gradient boosting model for better predictive performance
tensorflow / keras	For deep learning and time-series modeling (LSTM/GRU networks).
Streamlit	Used to build a lightweight, interactive web interface to display predictions.
joblib / pickle	For model serialization and saving/loading trained models.
datetime	For handling and processing date/time values in patient records.

Table 3: Python Packages and Libraries

User Interface:

- **Streamlit:** An open-source Python framework to build interactive dashboards and visualization tools. Used to present model predictions in a clean, user-friendly web interface.

Key Variables Used in the Project

Variable Name	Description	Importance
patient_id	Unique identifier for each patient.	Used for grouping and time-series segmentation.
Age, Sex	Demographic features.	Possible predictors of disease severity.
Date of Onset, Date of Discharge, Date of Test & Time of Test	Time features related to disease progression.	Used in trend analysis and time-lagged features.
Symptoms	Categorical/binary symptom data (e.g., fever, rash).	Key indicators of severity and disease stage.
Platelet (cells/cu.mm)	Target variable (numeric, time-series).	Main value to be forecasted.
Hemoglobin (gm/Dl), Red Blood Cell Count (millions/cu.mm), Hematocrit (Packed Cell Volume) (%)	Lab test results used as features.	Input predictors; may indicate internal bleeding or severity.

Table 4: Key variables used

Hardware Specifications

Component	Specification
Processor	12th Gen Intel(R) Core(TM) i5-12450H 2.00 GHz
RAM	16.0 GB (15.7 GB usable)
Operating System	Windows 11 Home Single Language
Development Environment	Jupyter Notebook, VS Code, Streamlit, Anaconda

Table 5: Hardware Specifications

The development of the platelet forecasting system relies heavily on Python's data science ecosystem and a robust machine learning workflow. The variables are carefully selected based on clinical relevance and data availability. The software is designed to run efficiently on mid-range hardware, making it accessible for most healthcare research or institutional settings. The modular design ensures future adaptability for integration with hospital management systems or mobile health apps.

Chapter 4: Model Details and Implementation

4.1 Model building

This section presents a comprehensive overview of the modelling techniques used for forecasting platelet counts in dengue patients. The modelling strategy involves understanding the data structure, selecting appropriate algorithms, engineering relevant features, and validating the predictive performance of the models.

Model 1: Random Forest Regressor

Algorithm Description

Random Forest is an ensemble learning method based on decision trees. It builds multiple trees on random subsets of the data and averages the predictions.

Mathematical Structure:

Given training data $D = \{(x_i, y_i)\}_{i=1}^{n}$, the random forest prediction is:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} f_t(x)$$

Where:

- T is the total number of trees.
- $f_t(x)$ is the prediction of the t-th decision tree.

Advantages:

- Handles non-linear relationships.
- Robust to outliers and missing values.
- Performs well with relatively small datasets.

Implementation:

- Library: sklearn.ensemble.RandomForestRegressor
- Key parameters: n_estimators=100, max_depth=10
- Train-test split: 80–20 by patient

Model 2: LSTM (Long Short-Term Memory Network)

Algorithm Description

LSTM is a type of recurrent neural network (RNN) suitable for sequence prediction. It is effective in capturing long-term dependencies in time-series data.

Mathematical Structure:

For each time step *t*:

$$f_t = \sigma(W_f \,.\, [h_{t-1}, x_t] \; + b_f)$$
$$i_t = \sigma(W_i \,.\, [h_{t-1}, x_t] \; + b_i)$$
$$\tilde{C}_t = tanh(W_c \,.\, [h_{t-1}, x_t] \; + b_c)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$
$$o_t = \sigma(W_o \,.\, [h_{t-1}, x_t] \; + b_o)$$
$$h_t = \; o_t * tanh(C_t)$$

Where:

- x_t: Input vector at time t
- h_t: Hidden state
- C_t: Cell state

Implementation:

- Library: tensorflow / keras
- Input shape: (n_days, n_features)
- Layers:
  - LSTM layer (units=64)
  - Dense layer (units=1)
- Optimizer: Adam
- Loss: MSE

Training Strategy:

- Sequence data per patient
- Normalize features (MinMaxScaler)
- Early stopping and dropout used to prevent overfitting

Model 3: XGBoost Regressor (XGBRegressor)

Algorithm Description

XGBoost (Extreme Gradient Boosting) is an advanced implementation of the gradient boosting algorithm. It builds additive models in a forward, stage-wise fashion and allows optimization of arbitrary differentiable loss functions. It is highly efficient, flexible, and accurate for structured/tabular data.

Mathematical Structure:

XGBoost minimizes the regularized loss function:

$$L(\phi) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

Where:

- $l(y_i, \hat{y}_i)$: Loss function (e.g., squared error)
- $f_k \in F$: Each f_k is a decision tree
- $\Omega(f) = \gamma T + \frac{1}{2}\lambda ||w||^2$: Regularization term for model complexity
- T: Number of leaves in the tree
- w: Leaf weights
- γ, λ: Regularization parameters

Implementation

- Library Used: xgboost.XGBRegressor
- Input Features:
  - Lagged platelet counts (t-1, t-2)
  - Symptoms (binary encoded)
  - Demographic variables (age, sex)
  - Lab values (haemoglobin, hematocrit, RBC)
  - Days since onset
- Hyperparameters:
  - n_estimators = 200
  - max_depth = 6

- learning_rate = 0.05
- subsample = 0.8
- colsample_bytree = 0.8
- reg_lambda = 1, reg_alpha = 0.5

- Loss Function: Mean Squared Error (MSE)
- Training Strategy:
  - Patient-wise split to avoid data leakage

Implementation Requirements

Requirement	Details
Python Version	3.10
Libraries	pandas, scikit-learn, keras, tensorflow, matplotlib
Runtime Environment	Jupyter Notebook / VS Code

Table 6: Implementation Requirements

4.2 Data Inputs

This section explains the nature, structure, and processing of input data fed into the machine learning models for predicting platelet counts in dengue patients. It includes an overview of the raw data, preprocessing pipeline, and how the model transforms this data to yield accurate predictions.

4.2.1. Nature of the Input Data

The dataset consists of structured clinical data collected from 493 dengue patients. Each row represents a test entry recorded for a specific patient at a specific time. The dataset includes both static and time-varying features.

Key Data Categories:

Category	Features
Demographic Data	Patient ID, Age, Sex
Symptom Data	Thrombocytopenia, Fever related, Weakness & Fatigue, etc. (binary flags)
Lab Test Data	Platelet Count, Hemoglobin, RBC Count, Hematocrit
Time Information	Date of Onset, Date of Test, Days Since Onset
Target Variable	Platelet Count (next day or multiple days ahead, depending on model)

Table 7: Key Data Categories

4.2.2. Input Features to the Model

The models were fed with numerical and categorical features, engineered to enhance predictive performance. Here's a breakdown:

Time Series Features:

- platelet_count_t-1, t-2, ..., up to t-n: Historical platelet values
- Rolling mean and standard deviation of past 2–3 days

Patient Metadata:

- age: Numeric
- sex: Binary encoded (0 = Female, 1 = Male)

Lab Values:

- Hemoglobin (gm/Dl): Float
- Red Blood Cell Count (millions/cu.mm): Float
- Hematocrit (Packed Cell Volume) (%): Float

Symptoms (binary-encoded):

- fever, rash, bleeding, vomiting, NS1 Ag positive, etc.

Temporal Info:

- days_since_onset: Integer
- date_of_test: Used for time alignment

4.2.3. Data Preprocessing

To prepare the data for modelling, the following steps were applied:

4.2.4. Data Pipeline Flow

1. **Raw Input**: Patient test records including date-wise lab results and symptom flags.

2. **Preprocessing Layer**:
   - Missing value imputation
   - Encoding and scaling
   - Feature creation (lags, rolling means, etc.)

3. **Model Input Structure**:
   - For LSTM: Sequences per patient → shape (n_timesteps, n_features)
   - For XGBoost / Random Forest: Flat tabular vectors → shape (n_samples, n_features)

4. **Prediction Output**:
   - Predicted platelet_count for next day (t+1), or optionally for t+1 to t+3

4.2.5. Model Output Interpretation

The model returns a predicted platelet count value based on the most recent information available. This value is clinically useful to:

- Track downward trends indicating risk of haemorrhage
- Plan interventions like hospitalization or transfusion
- Compare patient progress across different treatment plans

4.3 User interface

To make the platelet count prediction tool accessible and user-friendly, a web-based interface was developed using Streamlit, an open-source Python framework for deploying interactive machine learning applications. This interface allows users to upload or input patient data, view

visual analytics, and generate platelet count forecasts without requiring any additional software installation on the client machine. The app can be accessed through a web browser, making it highly convenient for clinical or academic use.

The layout of the interface is structured with a sidebar for user input and a main panel for displaying outputs:

- **Sidebar Features**:
  - Input fields for patient data: age, sex, date of onset, symptoms (via dropdown), lab test date, and lab values (Hemoglobin, Platelet count, RBC, Hematocrit).
  - A 'Predict' button that triggers data preprocessing, feature engineering (e.g., lag and delta features), and prediction.

- **Main Panel**:
  - Displays the predicted platelet counts for the next 3 to 5 days in tabular form.
  - Includes visualizations like a line plot showing platelet trends over time.

The app also incorporates custom styling for a dark-themed dashboard, improving readability and user experience. Background image is subtly faded during prediction output to reduce visual clutter. The symptoms column, which was initially a free-text field, is transformed into an intuitive multi-select dropdown, allowing consistent and user-friendly input.

This interactive platform not only supports data visualization and exploration but also integrates machine learning seamlessly into a clinical decision-support environment.

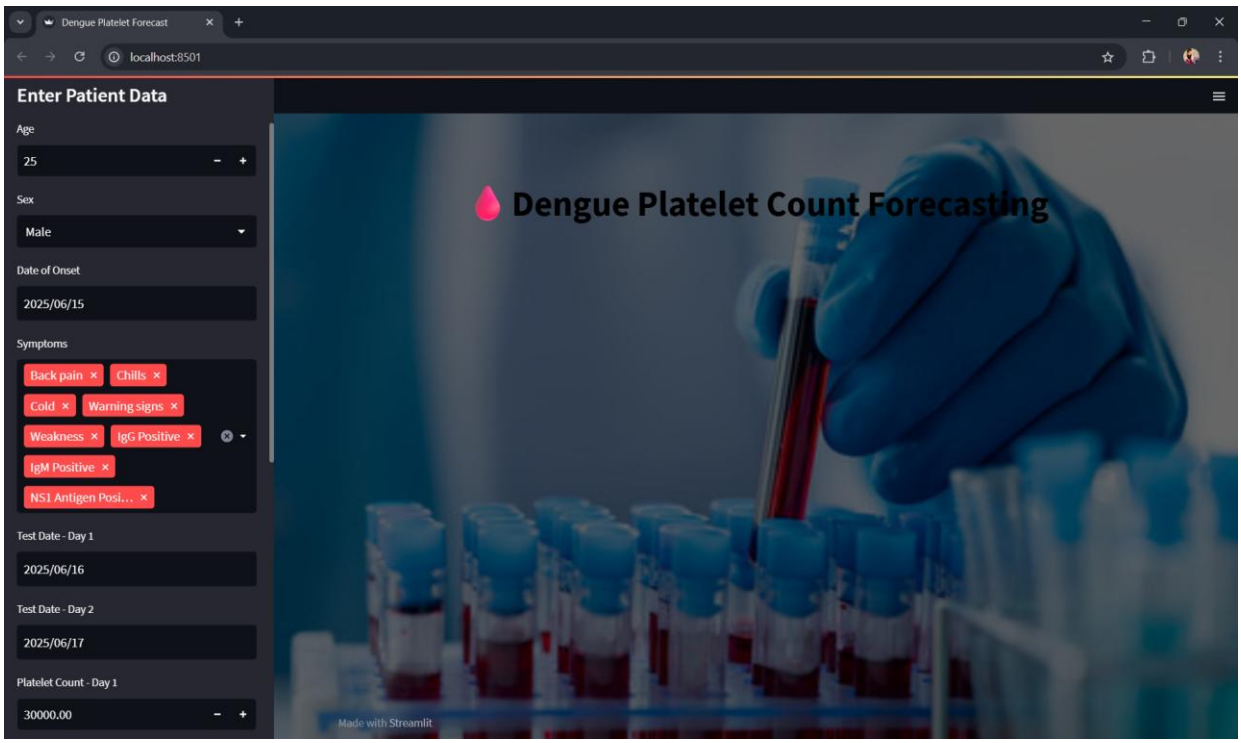
Below are screenshots of the interface:

Dengue Platelet Forecast
localhost:8501
Enter Patient Data
Age
0
Sex
Male
Date of Onset
2025/06/19
Symptoms
Choose an option
Test Date - Day 1
2025/06/19
Test Date - Day 2
2025/06/19
Platelet Count - Day 1
0.00
Platelet Count - Day 2
Dengue Platelet Count Forecasting
Made with Streamlit

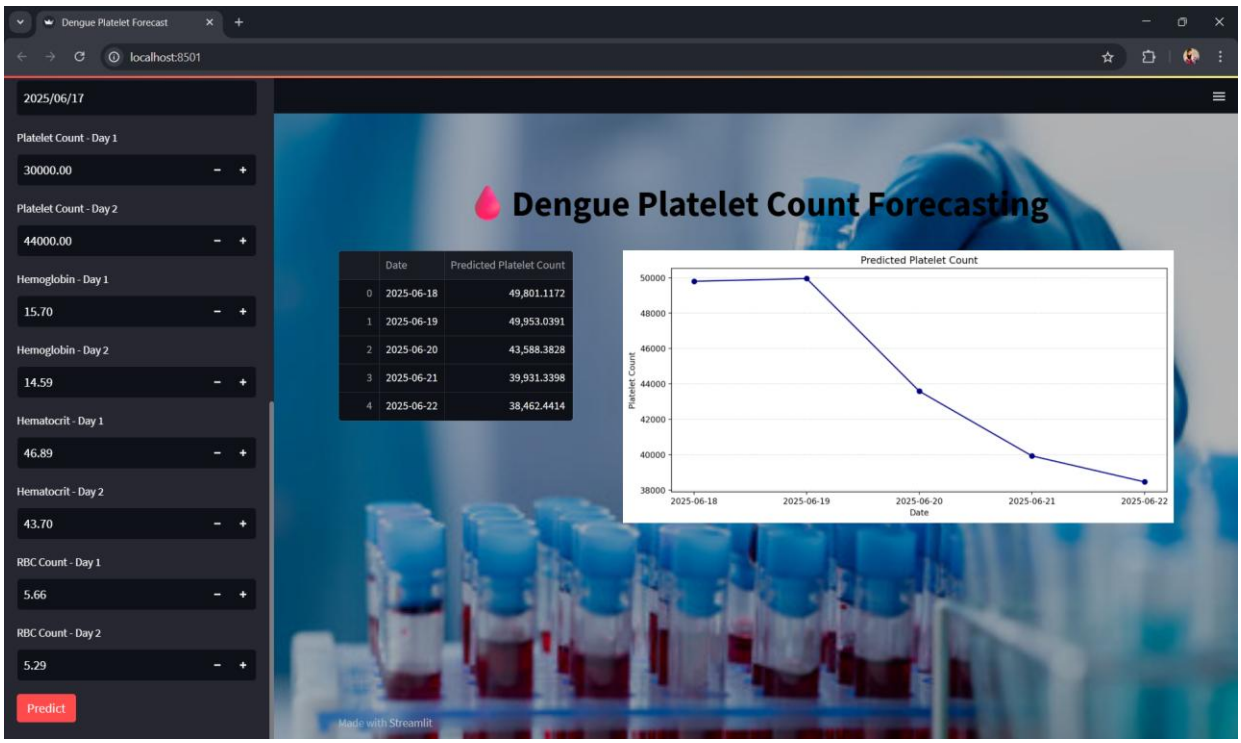
(a)

Dengue Platelet Forecast
localhost:8501
Enter Patient Data
Age
25
Sex
Male
Date of Onset
2025/06/19
Symptoms
Back pain
Bleeding
Blepharitis
Blood Disorders
Body pain
Chills
Clinical fluid accumulation
Cold
Conjunctival injection
Dengue Platelet Count Forecasting
Made with Streamlit

(b)



(c)



(d)

Fig 25: User Interface of the website

4.4 Results & Model Accuracy

After the development of the platelet prediction model using machine learning techniques (such as XGBoost, Random Forest and LSTM models), the system was rigorously evaluated to ensure its feasibility, reliability, and predictive accuracy. The model was trained on a cleaned and feature-engineered dataset of 493 dengue-positive patients, with the goal of forecasting platelet counts for the next 3 to 5 days based on early clinical indicators.

Model Evaluation Metrics

The predictive modelling phase of the project involved training and evaluating multiple machine learning algorithms to forecast platelet counts in dengue patients. The performance of each model was assessed using standard regression metrics, namely Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the R-squared (R^2) score. These metrics were chosen to capture both the absolute magnitude of error and the ability of the model to explain variability in the target variable — platelet count.

Among the models tested, the XGBoost Regressor (XGBRegressor) showed robust performance, achieving an average MAE of 12,875.19, RMSE of 41,974.13, and an R^2 score of 0.76. These results suggest that the model was relatively accurate in estimating the platelet trends while maintaining a good balance between bias and variance. XGBoost benefited significantly from the well-engineered features such as lag values, delta changes in platelet and hematocrit counts, and encoded symptom variables. Its gradient boosting architecture allowed it to model complex, non-linear relationships in the data, which is particularly important in clinical time-series prediction where patient responses can vary widely.

In contrast, the Long Short-Term Memory (LSTM) network — a deep learning model suited for time-series data — produced less favorable results, with an MAE of 69,104.47, RMSE of 93,450.42, and a very low R^2 score of 0.07. This indicates that the LSTM model struggled to capture the platelet progression trends in the given dataset. The comparatively poor performance of LSTM may be due to the relatively small sample size (per patient time-series),

which is often insufficient for training deep learning models that require large volumes of sequential data. Additionally, the variation in test intervals and sparsity in time-series data likely posed challenges for LSTM's memory mechanisms, which depend on regular temporal patterns.

The Random Forest Regressor, on the other hand, demonstrated strong predictive capability, outperforming both XGBoost and LSTM in certain metrics. It achieved an MAE of 14,658.77, RMSE of 33,371.79, and an R^2 score of 0.83, making it the best performer in terms of variance explanation. The ensemble nature of the Random Forest allowed it to handle feature interactions and outliers effectively, and it was less sensitive to noise in the data. Despite having a slightly higher MAE than XGBoost, its significantly lower RMSE and superior R^2 suggest it was more consistent and reliable in predicting sharp drops or rises in platelet counts — a critical factor in managing dengue progression.

Overall, the comparison highlights that tree-based ensemble methods (Random Forest and XGBoost) are more suitable for the nature of this clinical data compared to sequence-based deep learning methods like LSTM. These models offer greater transparency, interpretability, and resilience to the irregularities present in real-world medical datasets. Based on these findings, the Random Forest Regressor was identified as the most feasible model for deployment in the Streamlit-based user interface, followed closely by XGBoost for its balance between speed and accuracy.

4.5 Result Analysis & Discussion

The results obtained from the predictive models reflect both the challenges and strengths of modeling platelet progression in dengue patients using clinical and laboratory data. The variations in performance across models can be attributed to the nature of the dataset, the temporal characteristics of platelet count behavior, and the modeling approach used.

Among all the models tested, the Random Forest Regressor performed the best with an R^2 score of 0.83, indicating that the model could explain 83% of the variance in the platelet count values.

It also achieved a relatively low RMSE of 33,371.79 and MAE of 14,658.77, showing its ability to make precise and consistent predictions. This high performance can be explained by the ensemble nature of the Random Forest algorithm, which builds multiple decision trees and averages their outputs. This reduces overfitting and captures non-linear interactions between features such as lag values, delta changes, and symptoms — all of which are crucial for modeling a complex, fluctuating health indicator like platelet count.

The XGBoost Regressor also performed well with an R^2 of 0.76, MAE of 12,875.19, and RMSE of 41,974.13. While slightly less accurate than Random Forest in capturing extreme variations, XGBoost still demonstrated strong performance due to its gradient boosting mechanism, which sequentially corrects the errors of weak learners. XGBoost was especially effective in handling missing data and high-dimensional feature interactions, making it a suitable model for medical datasets with incomplete lab test results.

On the other hand, the LSTM model, which was expected to perform well due to its time-series structure, underperformed significantly with a very low R^2 score of 0.07, high MAE of 69,104.47, and RMSE of 93,450.42. This can be attributed to the limited volume of per-patient time-series data and the irregularity in test intervals. Deep learning models like LSTM typically require large and uniformly spaced temporal datasets to learn sequential patterns effectively. In this project, the sparse and inconsistent time steps across patients likely prevented the LSTM from learning meaningful trends. Additionally, LSTM is more sensitive to overfitting on small datasets, which may have further contributed to its poor generalization.

When compared to findings from the literature review, most existing dengue prediction studies have focused on binary classification tasks such as severity prediction or hospitalization risk using symptoms and static lab values. Few studies have tackled time-series forecasting of platelet counts, especially at the patient level. The models developed in this project — particularly the tree-based ensemble models — demonstrate that daily platelet progression can be modeled effectively using early-day data, which is a novel contribution in the context of dengue disease management.

In conclusion, the Random Forest Regressor emerged as the most reliable and clinically relevant model in this study. Its ability to handle missing data, capture non-linear trends, and provide interpretable outputs makes it suitable for real-world healthcare applications. The comparison also highlights that classical machine learning models, when supported by proper feature engineering, can outperform more complex deep learning models in low-data medical environments. These findings reinforce the importance of data quality, structure, and domain-specific feature design in building effective predictive systems for healthcare.

Chapter 5: Conclusion

5.1 Summary

This project focused on the development of a data-driven clinical decision support system for predicting platelet count progression in dengue patients, using early clinical data and laboratory results. Recognizing the critical importance of timely platelet monitoring in dengue management, the system aims to assist healthcare professionals by forecasting platelet trends for the upcoming days — particularly during the early stages of the illness when interventions are most crucial.

The project began with comprehensive data preprocessing, where raw clinical data collected from Fr. Muller's Medical College Hospital was cleaned, standardized, and transformed into a structured dataset. This included handling missing lab results, splitting free-text symptom entries into structured features, engineering time-based variables like lag features and days since onset, and ensuring each patient record was properly indexed and sorted. The innovation here lies in the automated transformation of unstructured medical data into machine learning-ready format, allowing the models to extract meaningful patterns from complex clinical timelines.

A comparative modeling approach was applied, involving XGBoost Regressor, Random Forest Regressor, and LSTM models. The Random Forest Regressor emerged as the most effective, achieving an R^2 score of 0.83 and low error metrics, making it a practical choice for deployment. Unlike many existing studies that focus only on disease classification, this project introduced a time-series regression approach to dengue progression prediction — an area that has received limited attention in research.

One of the standout features of this project is the creation of an interactive, no-installation web interface using Streamlit, which allows users to input patient data and instantly visualize predicted platelet trends. This adds real-world usability to the model, making it accessible for doctors, students, and researchers without requiring any specialized software.

In conclusion, the project combines medical relevance, machine learning, and user-centered design to offer a complete and innovative solution to dengue patient monitoring. Its key achievements include effective data transformation, accurate short-term forecasting of platelet levels, and the development of an intuitive web interface — making it a pioneering effort in predictive analytics for tropical diseases.

5.2 Limitations

While the developed system for predicting platelet counts in dengue patients has shown promising results and practical utility, it is not without limitations. One of the primary constraints is the limited size and diversity of the dataset. The data used for model training was collected from a single institution — Fr. Muller's Medical College Hospital — and included 493 patients. Although the dataset was rich in clinical detail, its relatively small scale and regional concentration may limit the generalizability of the model to broader populations across different geographical locations, age groups, or clinical settings.

Another limitation lies in the irregularity and sparsity of time-series lab tests. Since platelet and hematocrit readings were not recorded at uniform intervals for all patients, time-based models such as LSTM underperformed. This inconsistency also posed challenges during feature engineering and may have affected the accuracy of the lag and delta features. A more structured and consistent temporal dataset would allow for deeper time-series modeling and improved performance of sequence-based algorithms.

Additionally, the current system does not include certain clinical variables that could further enhance predictive accuracy, such as white blood cell count, liver enzyme levels, comorbidities, or medication history. The model also assumes that early lab values and symptoms alone can predict platelet trends, which may not capture more complex or atypical cases.

From a deployment perspective, while the Streamlit interface is accessible and user-friendly, it remains a prototype-level tool that would require integration with hospital information systems (HIS) and electronic medical records (EMR) to be used in real clinical environments. Features like user authentication, patient data storage, real-time updates, and doctor-specific dashboards are yet to be implemented.

These limitations highlight the scope for future enhancements, including expanding the dataset across multiple hospitals, incorporating more comprehensive clinical features, exploring deep learning models with attention mechanisms, and developing a fully integrated clinical tool. Addressing these challenges will not only improve the accuracy and robustness of the model but also move the system closer to real-world clinical deployment.

5.3 Future scope

The current system presents a significant step toward data-driven monitoring of dengue patients through platelet count prediction. However, there is ample scope for future development that can enhance its clinical relevance, accuracy, scalability, and integration into real-world healthcare settings.

One of the most impactful future enhancements would be the expansion of the dataset, both in volume and diversity. Collecting data from multiple hospitals across different regions would help the model learn from a wider variety of patient profiles and clinical conditions, making predictions more generalizable. A larger dataset would also allow the application of more advanced deep learning models, such as transformer-based architectures or attention-equipped LSTMs, which require substantial data to perform optimally.

Another important direction is the inclusion of additional clinical features such as white blood cell count (WBC), liver function tests (LFTs), vital signs, comorbidities, and treatment history. These variables can offer deeper insights into disease severity and trajectory, improving the model's ability to capture complex interactions that influence platelet dynamics.

On the deployment side, the Streamlit interface can be upgraded into a more comprehensive, secure, and real-time clinical decision support tool. This could involve:

- Integration with hospital EMR/HIS systems
- Secure login and role-based access for clinicians
- Automatic data upload and real-time prediction

- Patient-wise dashboards for trend monitoring
- Mobile-friendly interface for accessibility

Furthermore, the system could be extended to predict additional clinical risks, such as the likelihood of bleeding complications or hospitalization needs, making it a more holistic decision-support tool for dengue case management. Real-time alert systems for doctors based on prediction thresholds could also be developed.

Overall, by extending its data sources, refining its features, and improving its deployment pipeline, this system has the potential to evolve from a research prototype into a clinically reliable tool for early dengue risk stratification and personalized patient care.

References

[1] K. Phakhounthong, P. Chaovalit, P. Jittamala, S. D. Blacksell, M. J. Carter, P. Turner, K. Chheng, S. Sona, V. Kumar, N. P. J. Day, L. J. White, and W. Pan-ngum, "Predicting the severity of dengue fever in children on admission based on clinical features and laboratory indicators: application of classification tree analysis," presented at BMC Pediatrics, 2018.

[2] I. Ghosh, S. Gupta, and S. Rana, "Anticipating dengue outbreaks using a novel hybrid ARIMA-ARNN model with exogenous variables," presented at arXiv preprint arXiv:2312.15270, Dec. 23, 2023.

[3] I. N. Faridah, H. Dania, Y.-H. Chen, W. Supadmi, B. D. Purwanto, M. J. Heriyanto, M. A. Aufa, W.-C. Chang, and D. A. Perwitasari, "Dynamic changes of platelet and factors related dengue haemorrhagic fever: A retrospective study in Indonesian," presented at Diagnostics (Basel), Apr. 11, 2022.

[4] L. Tanner, M. Schreiber, J. G. H. Low, A. Ong, T. Tolfvenstam, Y. L. Lai, L. C. Ng, Y. S. Leo, L. T. Puong, S. G. Vasudevan, C. P. Simmons, M. L. Hibberd, and E. E. Ooi, "Decision tree algorithms predict the diagnosis and outcome of dengue fever in the early phase of illness," presented at PLoS Neglected Tropical Diseases, Mar. 12, 2008.

[5] H. Mayrose, G. M. Bairy, N. Sampathila, S. Belurkar, and K. Saravu, "Machine learning-based detection of dengue from blood smear images utilizing platelet and lymphocyte characteristics," presented at Diagnostics, Jan. 6, 2023.

[6] M. Panja, T. Chakraborty, S. S. Nadim, I. Ghosh, and U. Kumar, "An ensemble neural network approach to forecast dengue outbreak based on climatic condition," presented at arXiv preprint arXiv:2212.08323v2, Dec. 20, 2022.

[7] N. T. Tran, J. Xin, and G. Zhou, "FWin transformer for dengue prediction under climate and ocean influence," presented at arXiv preprint arXiv:2403.07027, Mar. 10, 2024.

[8] Q. He, Z. You, Q. Dong, J. Guo, and Z. Zhang, "Machine learning for identifying risk of death in patients with severe fever with thrombocytopenia syndrome," presented at the Frontiers in Microbiology, Sep. 13, 2024.

[9] S. Q. Ong, P. Isawasan, A. M. M. Ngesom, H. Shahar, A. M. Lasim, and G. Nair, "Predicting dengue transmission rates by comparing different machine learning models with vector indices and meteorological data," presented at Scientific Reports, 2023.

[10] P. Silitonga, A. Bustamam, H. Muradi, W. Mangunwardoyo, and B. E. Dewi, "Comparison of dengue predictive models developed using artificial neural network and discriminant analysis with small dataset," presented at Applied Sciences, Jan. 21, 2021.

[11] G. M. Hair, F. F. Nobre, and P. Brasil, "Characterization of clinical patterns of dengue patients using an unsupervised machine learning approach," presented at BMC Infectious Diseases, 2019.